Sinais e Sistemas para Engenharia

MÓDULO 1B – SINAIS ELEMENTARES

VERSÃO OCTAVE

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Sinais Elementares:

1) DEGRAU UNITÁRIO
2) SIGNUM
3) RAMPA
4) PULSO
5) PULSO ÁREA UNITÁRIA
6) IMPULSO CONTÍNUO
7) IMPULSO DISCRETO
8) PULSO TRIANGULAR
9) PULSO GUASSIANO
10) SINC
11) PULSO SINC
12) SINAL EXPONENCIAL REAL
13) PULSO EXPONENCIAL
14) FASOR $\exp(\%i * w * t)$
15) SINAL EXPONENCIAL COMPLEXO $\exp\big((\sigma + \%i * w) * t\big)$
16) SENÓIDES $\sin(w * t)$
17) SENÓIDES DISCRETAS $\exp(\%i * w * \frac{n}{N})$

```
% Degrau unitario
function  x=f_u(t)
    x =double(t>=0);
end;

% Signum
function  x=f_s(t)
    x =2*f_u(t)-1;
end;

% Funcao Rampa
function  x=f_r(t)
    x =t.*f_u(t);
end;

%pulso
function  x=f_p(t,T)
    x =f_u(t+T/2)-f_u(t-T/2);
end;

%Pulso Area Unitaria largura T
function  x=f_pd(t,T)
    x =1/T*f_p(t,T);
end;

% Impulso Discreto
function  x=f_d(n)
    x = double(n==0);
end;

%Pulso triangular largura T
function  x=f_pt(t,T)
    x =(1-2/T*abs(t)).*f_p(t,T);
end;

% sinc sinais e sistemas
function  x=f_sc(t,T)
    x =sinc(t);
end;

% Pulso Sinc Periodo T
function  x=f_psc(t,T)
    x =1/T*sinc(t/T);
end;

 % Pulso Gaussiano (med,std)
function  x=f_pg(t,med_in,std_in)
    x =1/(std_in*sqrt(2*pi))*exp(-1/2*((t-med_in)/std_in).^2 );
end;

 % Sinal Exponencial
function  x=f_exp(t,a)
    x = exp(a*t);
end;

 % Pulso Exponencial
function  x=f_pexp(t,T)
    x =1/(2*T)*exp(-abs(t)/T);
end;
```

Sinais Elementares:

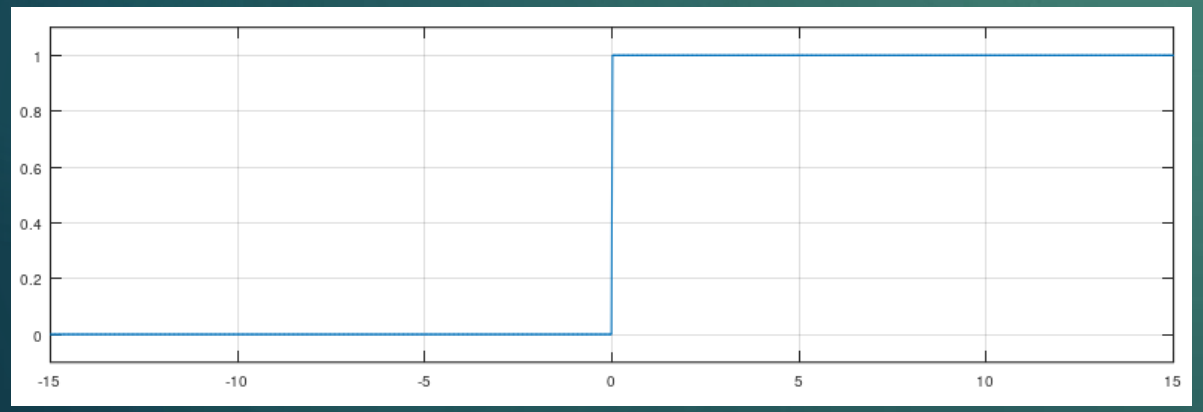
1) Degrau Unitário

```
% Degrau unitario
function  x=f_u(t)
    x =double(t>=0);
end;
```

```
% Degrau unitario
f_u= @(t) double(t>=0);
```

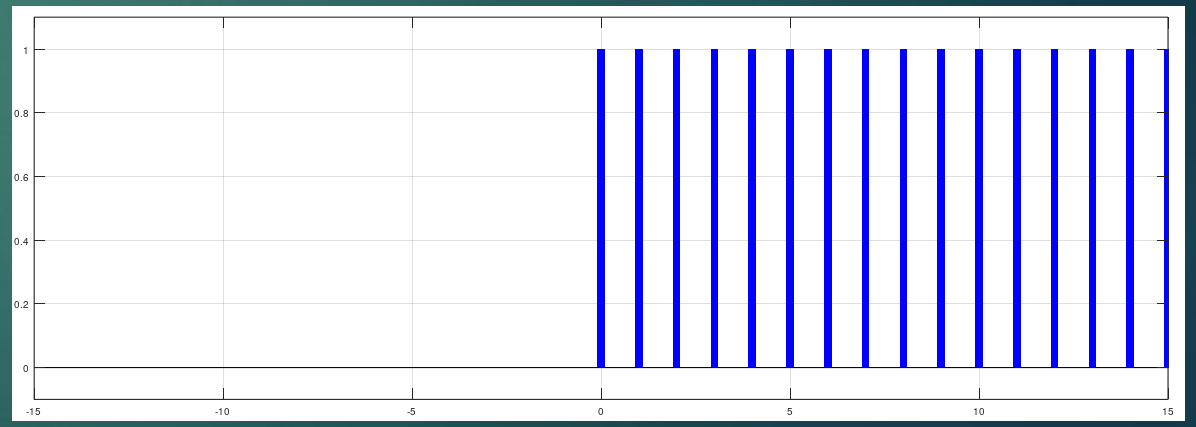
Degrau Contínuo $u(t) = \begin{cases} 0; t < 0 \\ 1; t \geq 0 \end{cases}$

```
>> t=linspace(-15,15,1000);
>> x=f_u(t);
>> plot(t,x);
>> grid on;
>> axis([-15,15,-0.1,1.1])
```



Degrau Discreto $u[n] = \begin{cases} 0; n < 0 \\ 1; n \geq 0 \end{cases}$

```
>> n=[-15:15];
>> xn=f_u(n);
>> bar(n,xn,0.2,'b','edgecolor','none');
>> grid on;
>> axis([min(n),max(n),-0.1,1.1])
```



2) Signum

```
% Signum
function  x=f_s(t)
    x =2*f_u(t)-1;
end;
```

```
% Signum
f_s= @(t) (2*f_u(t)-1);
```

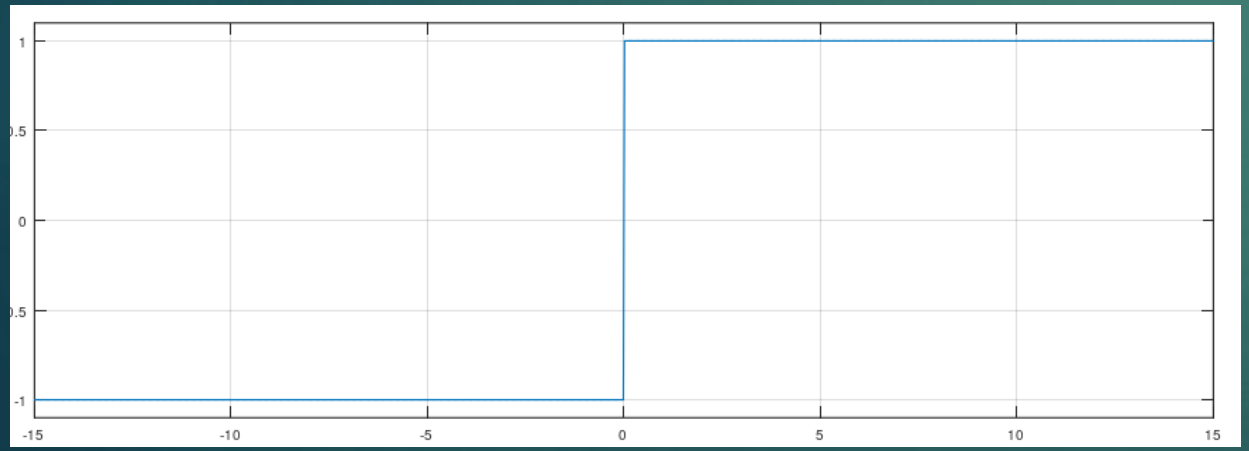
Contínua $\operatorname{sgn}(t) = \begin{cases} -1; t < 0 \\ 1\,; t \geq 0 \end{cases}$

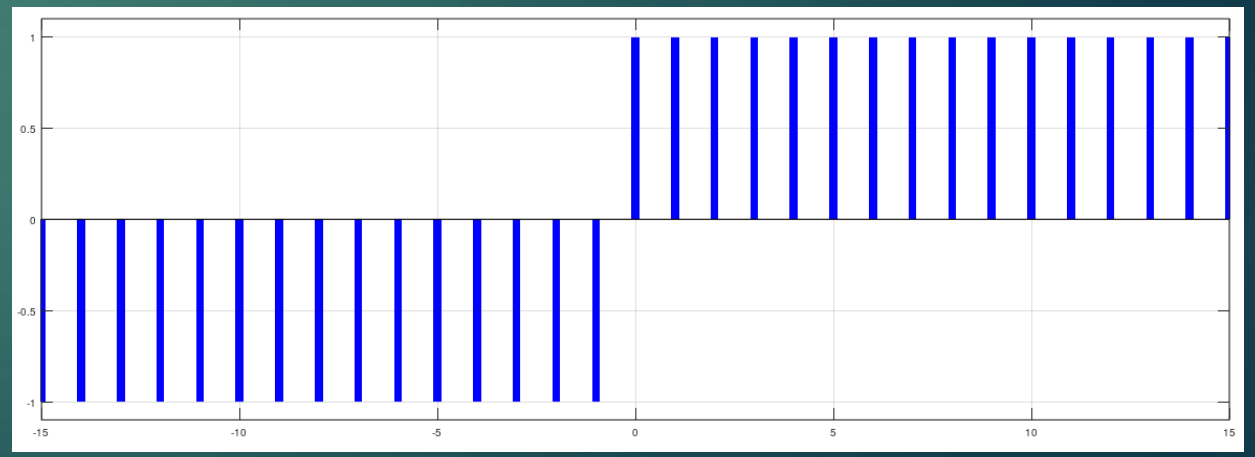
$$\operatorname{sgn}(t) = 2\mathrm{u}(\mathrm{t}) - 1$$

Discreta $\operatorname{sgn}[n] = \begin{cases} -1; n < 0 \\ 1; n \geq 0 \end{cases}$

```
>> t=linspace(-15,15,1000);
>> x=f_s(t);
>> plot(t,x);
>> grid on;
>> axis([-15,15,-1.1,1.1])
```

```
>> n=[-15:15];
>> xn=f_s(n);
>> bar(n,xn,0.2,'b','edgecolor','none');
>> grid on;
>> axis([min(n),max(n),-1.1,1.1])
```





3) Rampa

```
% Funcao Rampa
function  x=f_r(t)
    x =t.*f_u(t);
end;
```

```
% Funcao Rampa
f_r= @(t) t.*f_u(t);
```

Rampa Contínua $r(t) = \begin{cases} 0; t < 0 \\ t; t \geq 0 \end{cases}$

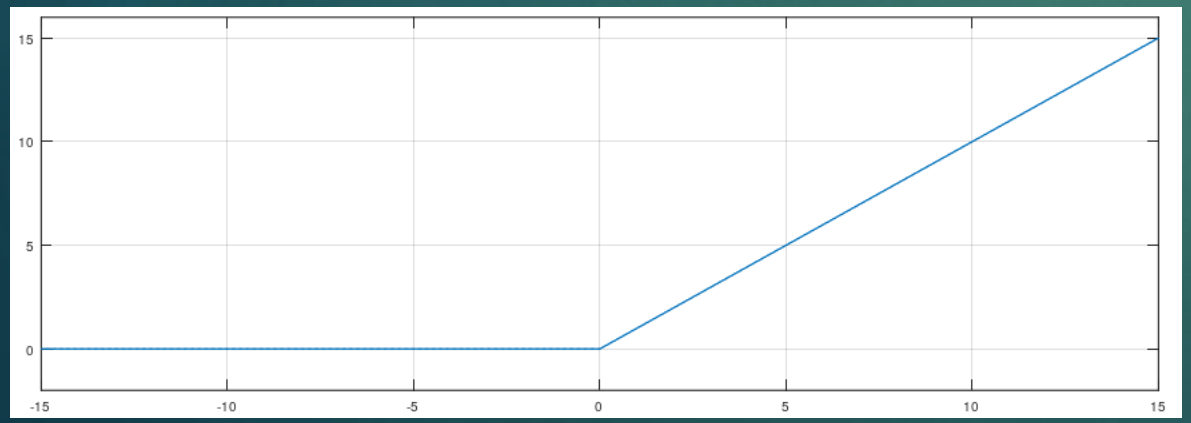
$r(t) = t\ u(t)$

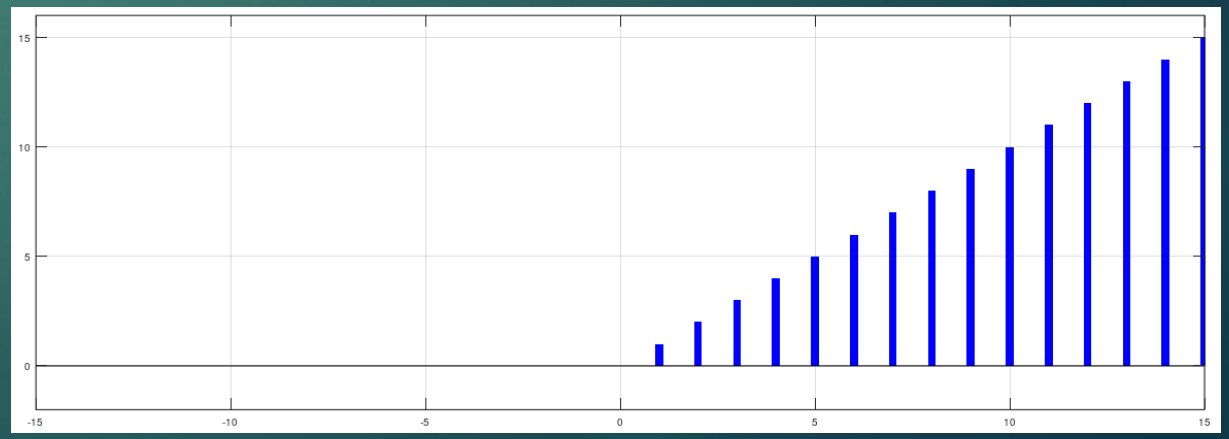
Rampa Discreta $r[n] = \begin{cases} 0; n < 0 \\ n; n \geq 0 \end{cases}$

$r[n] = n\ u[n]$

```
>> t=linspace(-15,15,1000);
>> x=f_r(t);
>> plot(t,x);
>> grid on;
>> axis([-15,15,-2,16])
>>
```

```
>> n=[-15:15];
>> xn=f_r(n);
>> bar(n,xn,0.2,'b','edgecolor','none');
>> grid on;
>> axis([min(n),max(n),-2,16])
>>
```





4) Pulso Largura T

Pulso contínuo: $p_T(t) = \begin{cases} 0; & t < -T/2 \\ 1; & -\frac{T}{2} \le t \le T/2 \\ 0; & t > T/2 \end{cases}$

```
%pulso
function  x=f_p(t,T)
    x =f_u(t+T/2)-f_u(t-T/2);
end;
```

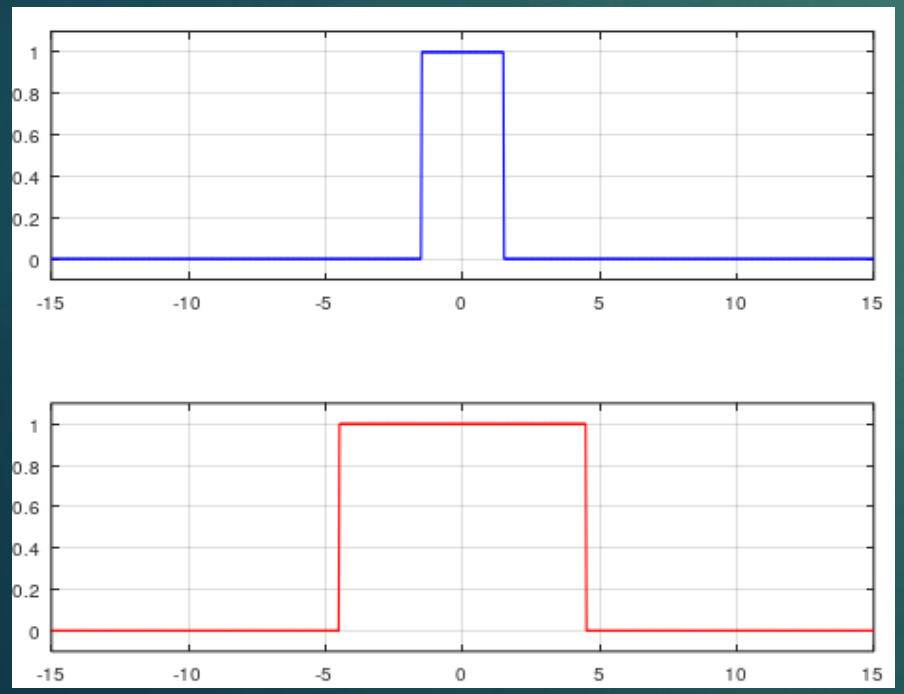
```
%pulso
f_p= @(t,T) f_u(t+T/2)-f_u(t-T/2);
```

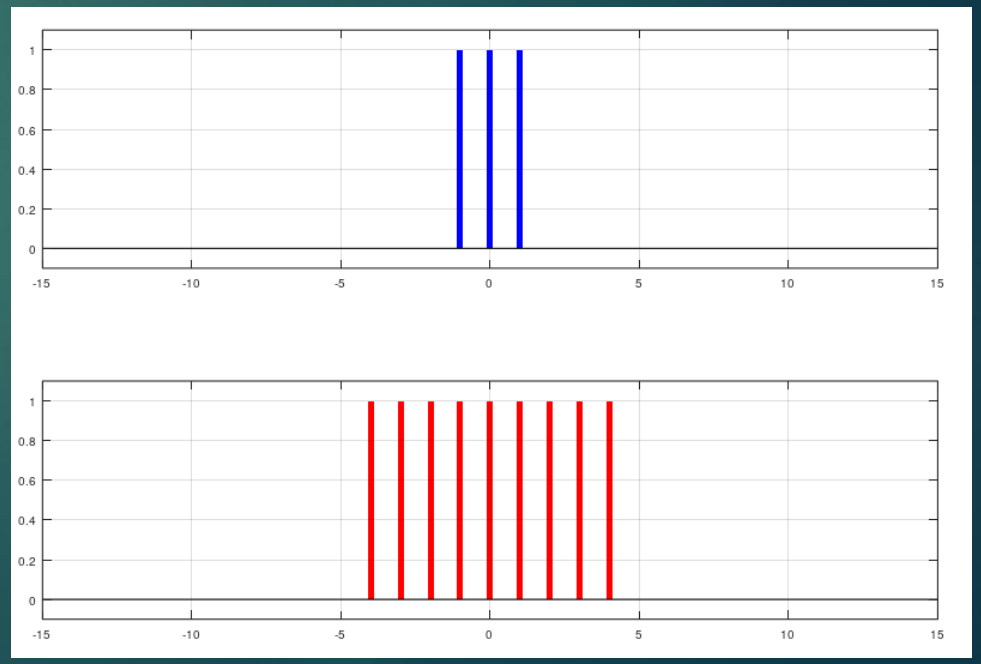
Pulso Discreto $p_T[n] = \begin{cases} 0; & n < -T/2 \\ 1; & -\frac{T}{2} \le n \le T/2 \\ 0; & n > T/2 \end{cases}$

```
>> t=linspace(-15,15,1000);
>> subplot(211)
>> x=f_p(t,3);
>> plot(t,x,'b');
>> grid on;
>> axis([-15,15,-0.1,1.1])
>> subplot(212)
>> x=f_p(t,9);
>> plot(t,x,'r');
>> grid on;
>> axis([-15,15,-0.1,1.1])
```

$$p_T(t) = u\left(t + \frac{T}{2}\right) - u\left(t - \frac{T}{2}\right)$$

```
>> n=[-15:15];
>> subplot(211)
>> xn=f_p(n,3);
>> bar(n,xn,0.2,'b','edgecolor','none');
>> grid on;
>> axis([min(n),max(n),-0.1,1.1])
>> subplot(212)
>> xn=f_p(n,9);
>> bar(n,xn,0.2,'r','edgecolor','none');
>> grid on;
>> axis([min(n),max(n),-0.1,1.1])
```



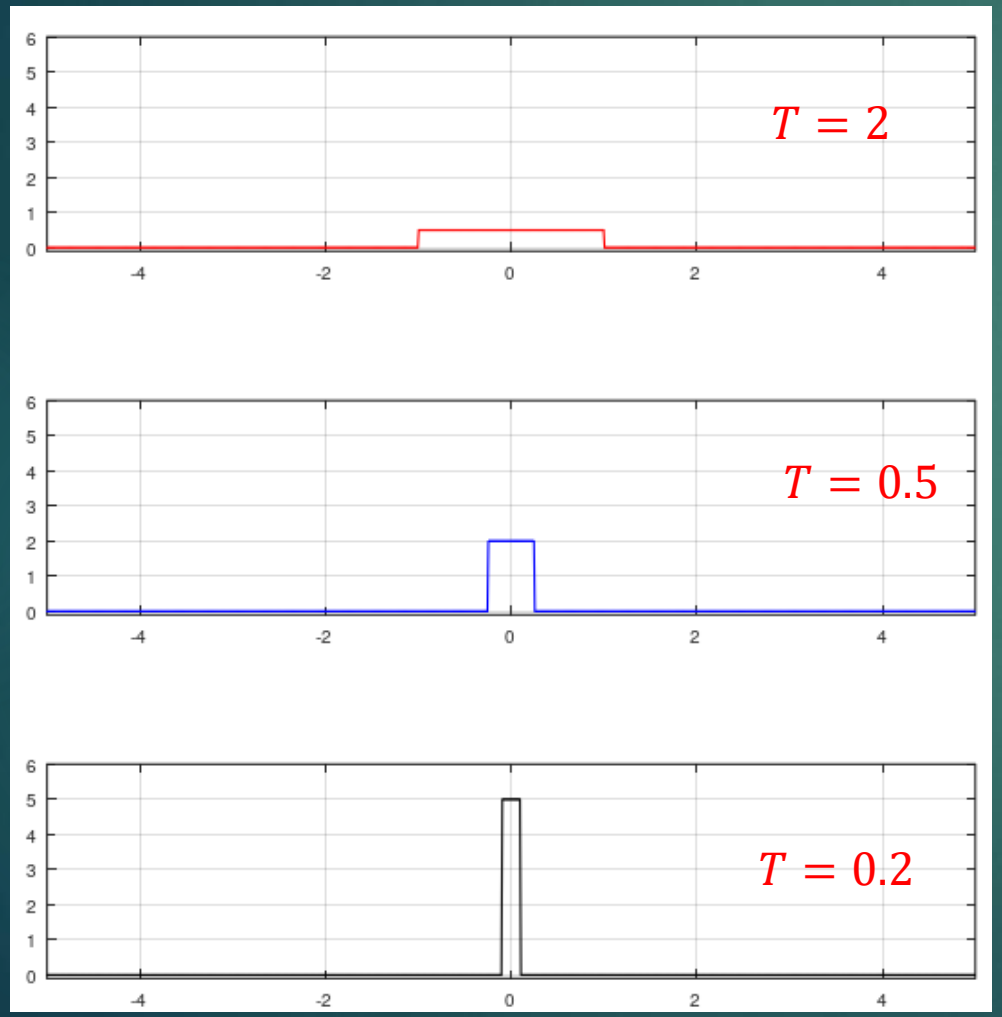


5) Pulso Largura T com área unitária

Pulso contínuo: $p_{Td}(t) = \begin{cases} 0; & t < -T/2 \\ 1/T; & -\frac{T}{2} \leq t \leq T/2 \\ 0; & t > T/2 \end{cases}$

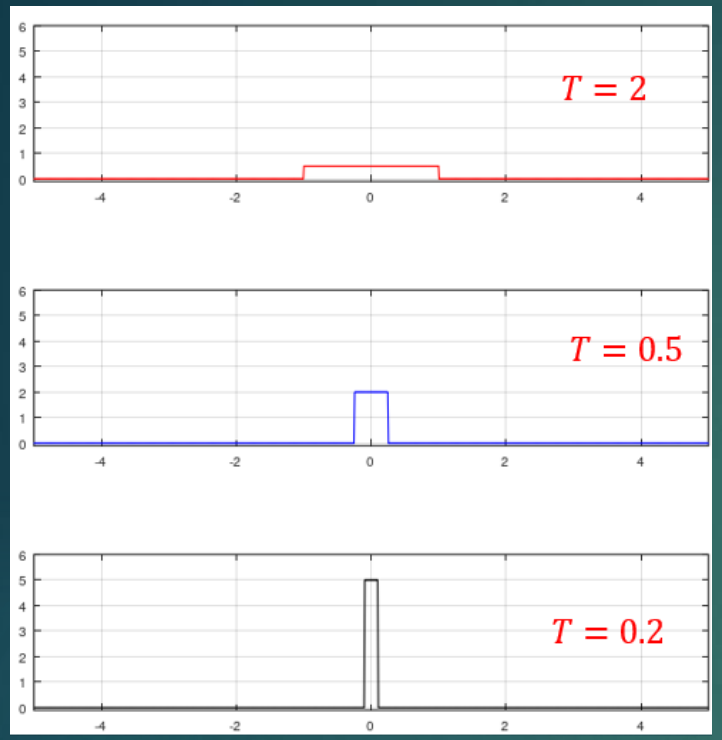
```
%Pulso Area Unitaria largura T
function  x=f_pd(t,T)
    x =1/T*f_p(t,T);
end;
```

```
%Pulso Area Unitaria largura T
f_pd= @(t,T) 1/T*f_p(t,T);
```




```
>> subplot(311)
>> x=f_pd(t,2);
>> plot(t,x,'r');
>> grid on;
>> axis([-5,5,-0.1,6])
>> subplot(312)
>> x=f_pd(t,0.5);
>> plot(t,x,'b');
>> grid on;
>> axis([-5,5,-0.1,6])
>> subplot(313)
>> x=f_pd(t,0.2);
>> plot(t,x,'k');
>> grid on;
>> axis([-5,5,-0.1,6])
```

6) Impulso Contínuo



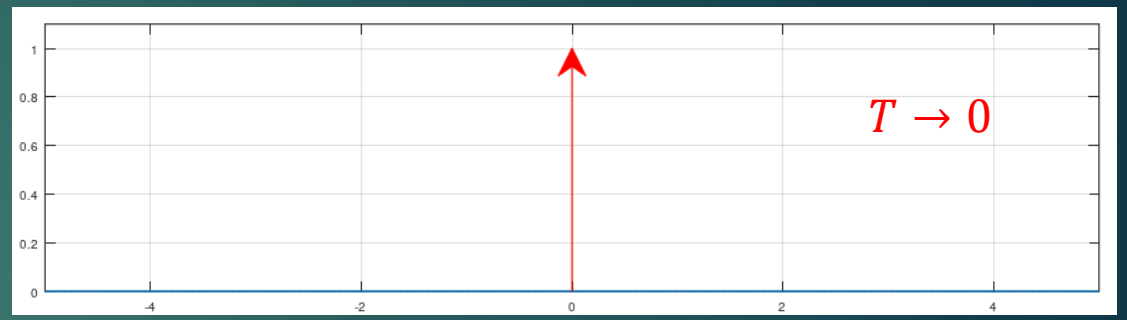
- Impulso Contínuo

$$\delta(t) = \begin{cases} 0; & t \neq 0 \\ \infty; & t = 0 \end{cases}$$

$$\int_{-\infty}^{+\infty} \delta(t)dt = \int_{0^-}^{0^+} \delta(t)dt = 1$$

$$\delta(t) = \lim_{T \to 0} p_{Td}(t)$$

```
>> t=linspace(-5,5,1001);
>> x=t*0;
>> plot(t,x);
>> grid on;
>> axis([-5,5,0,1.1])
>> plot_arrow_fga([0,0],[0,1],'r')
```




```
% Impulso Discreto
function  x=f_d(n)
    x = double(n==0);
end;
```

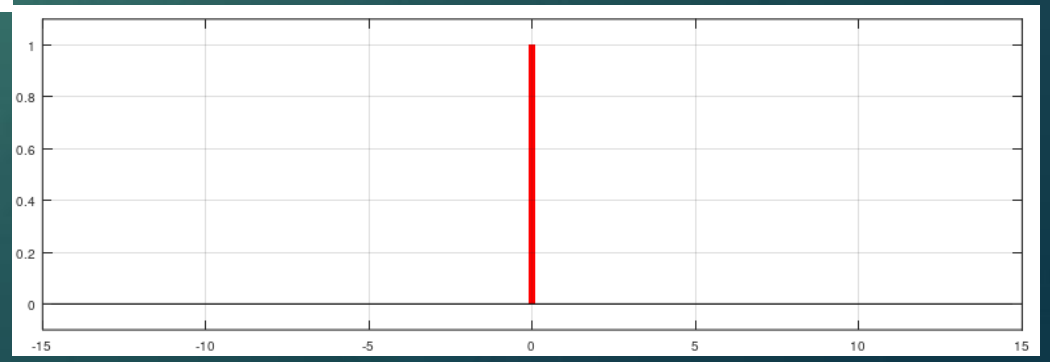
```
% Impulso Discreto
f_d= @(n) double(n==0);
```

```
>> n=[-15:15];
>> xn=f_d(n);
>> bar(n,xn,0.2,'r','edgecolor','none');
>> grid on;
>> axis([min(n),max(n),-0.1,1.1])
```

7) Impulso Discreto

Impulso Discreto

$$\delta[n] = \begin{cases} 1; & n = 0 \\ 0; & n \neq 0 \end{cases}$$



8) Pulso Triangular Largura T

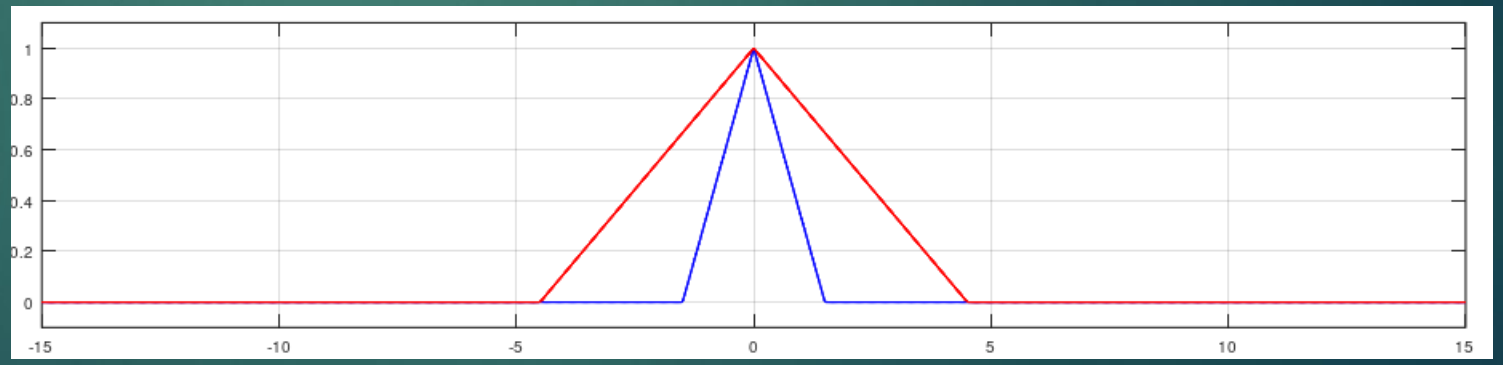
```
%Pulso triangular largura T
function  x=f_pt(t,T)
    x =(1-2/T*abs(t)).*f_p(t,T);
end;
```

```
%Pulso triangular largura T
f_pt= @(t,T) (1-2/T*abs(t)).*f_p(t,T);
```

Contínuo: $p_T(t) = \begin{cases} 0; & t < -T/2 \\ 1 - \frac{2}{T}|t|; & -\frac{T}{2} \leq t \leq T/2 \\ 0; & t > T/2 \end{cases}$

Discreto : $p_T[n] = \begin{cases} 0; & n < -T/2 \\ 1 - \frac{2}{T}|n|; & -\frac{T}{2} \leq n \leq T/2 \\ 0; & n > T/2 \end{cases}$

```
>> t=linspace(-15,15,1000);
>> hold on; grid on;
>> x=f_pt(t,3);
>> plot(t,x,'b');
>> x=f_pt(t,9);
>> plot(t,x,'r');
>> axis([-15,15,-0.1,1.1])
>>
```



9) Pulso Gaussiano (med,std)

```
% Pulso Gaussiano (med,std)
function  x=f_pg(t,med_in,std_in)
   x =1/(std_in*sqrt(2*pi))*exp(-1/2*((t-med_in)/std_in).^2 );
end;
```

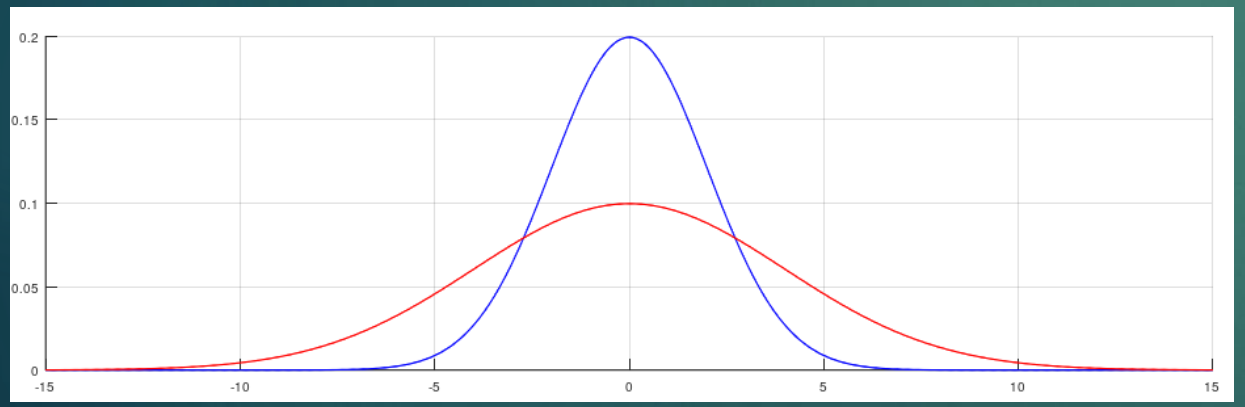
```
% Pulso Gaussiano (med,std)
f_pg= @(t,med_in,std_in) 1/(std_in*sqrt(2*pi))*exp(-1/2*((t-med_in)/std_in).^2 );
```

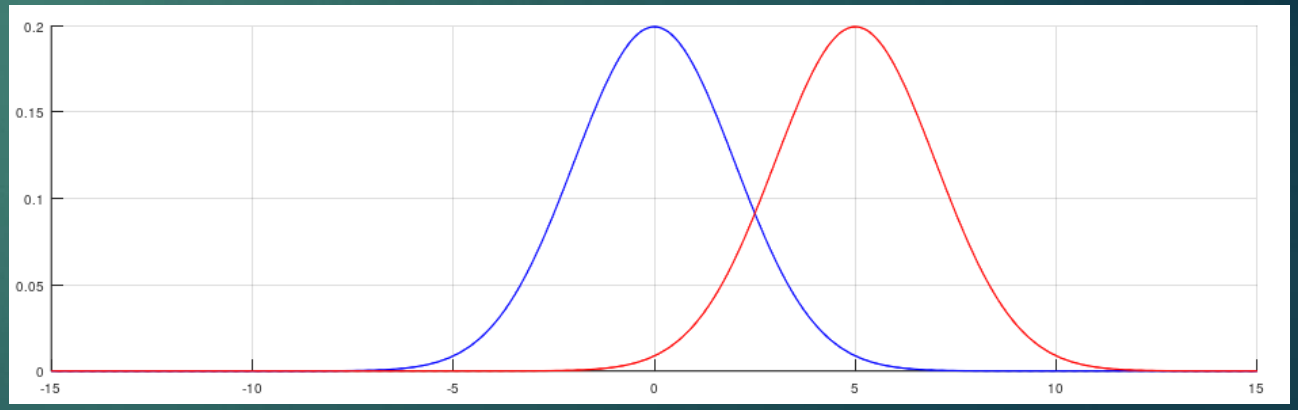
Contínuo: $p_{m,s}(t) = \frac{1}{s\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{t-m}{s}\right)^2}$

Discreto: $p_{m,s}[n] = \frac{1}{s\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{n-m}{s}\right)^2}$

```
>> t=linspace(-15,15,1000);
>> hold on; grid on;
>> x=f_pg(t,0,2);
>> plot(t,x,'b');
>> x=f_pg(t,0,4);
>> plot(t,x,'r');
>> axis([-15,15,-0.1,1.1])
```

```
>> t=linspace(-15,15,1000);
>> hold on; grid on;
>> x=f_pg(t,0,2);
>> plot(t,x,'b');
>> x=f_pg(t,5,2);
>> plot(t,x,'r');
```





10) Sinc

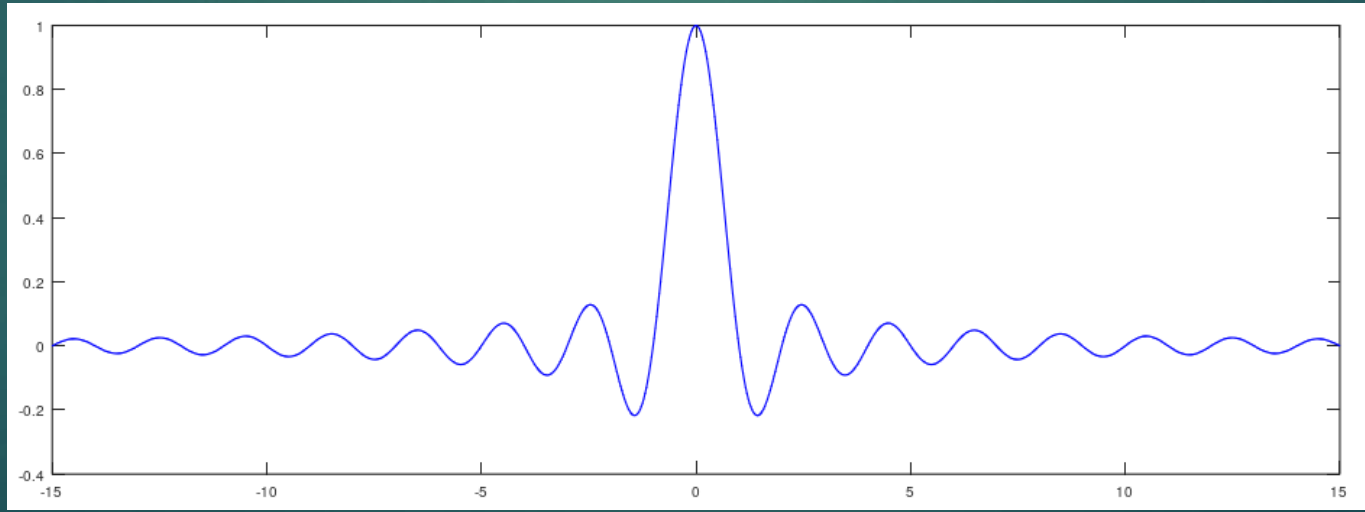
```
% sinc sinais e sistemas
function  x=f_sc(t)
    x =sinc(t);
end;
```

```
% sinc sinais e sistemas
f_sc= @(t) sinc(t);;
```

Contínuo: $x(t) = \frac{\sin(t)}{t}$

Discreto : $x[n] = \frac{\sin(n)}{n}$

```
>> t=linspace(-15,15,1000);
>> x=f_sc(t);
>> plot(t,x,'b');
>> grid on;
```



11) Pulso Sinc área unitária largura T

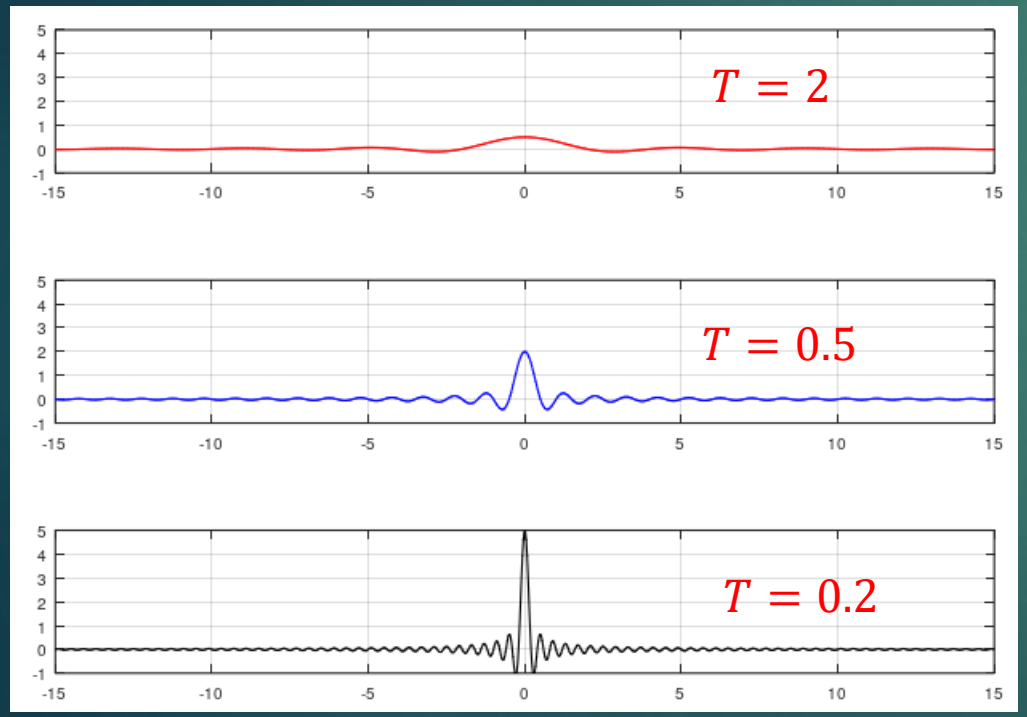
Contínuo $x_T(t) = \frac{1}{T}\frac{\sin\left(\frac{\pi t}{T}\right)}{\frac{\pi t}{T}}$

Discreto $x_T[n] = \frac{1}{T}\frac{\sin\left(\frac{\pi n}{T}\right)}{\frac{\pi t}{T}}$

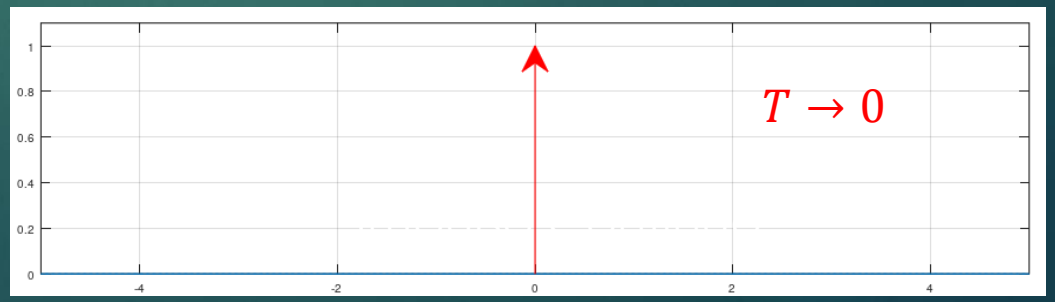
```
% Pulso Sinc Periodo T
function  x=f_psc(t,T)
    x =1/T*sinc(t/T);
end;
```

```
% Pulso Sinc Periodo T
f_psc= @(t,T) 1/T*sinc(t/T);
```

```
>> t=linspace(-15,15,1000);
>> subplot(311)
>> x=f_psc(t,2);
>> plot(t,x,'r');
>> grid on;
>> subplot(312)
>> x=f_psc(t,0.5);
>> plot(t,x,'b');
>> grid on;
>> subplot(313)
>> x=f_psc(t,0.2);
>> plot(t,x,'k');
>> grid on;
```



$$\delta(t) = \lim_{T\to 0} psc(t)$$



12) Sinal Exponencial Real

Contínua $x(t) = e^{at}$

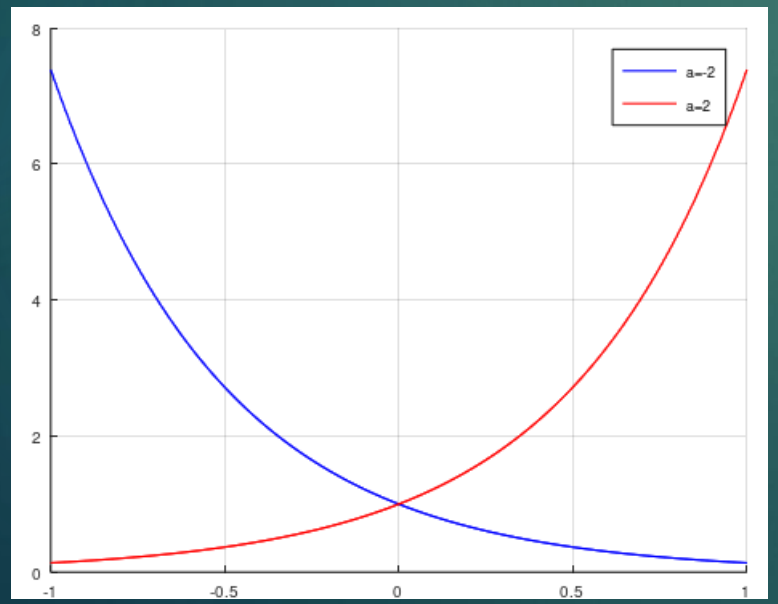
Discreta $x[n] = e^{an}$

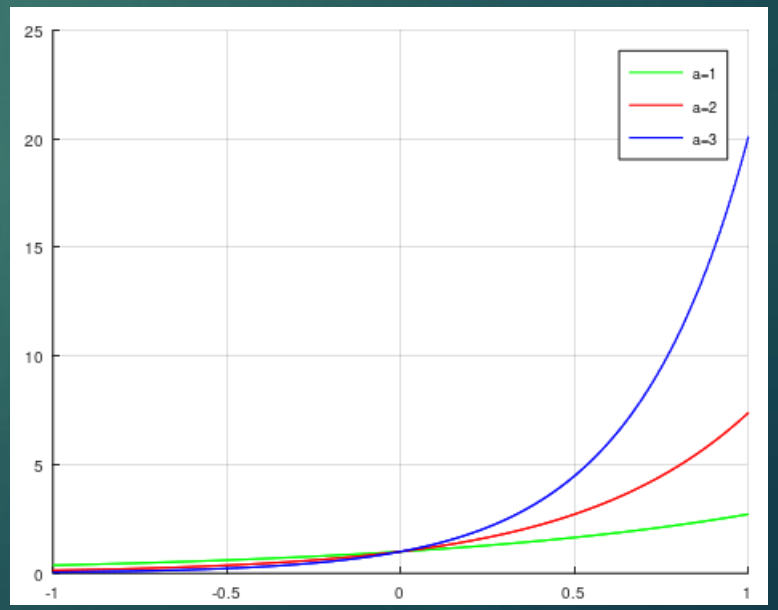
```
% Sinal Exponencial
function  x=f_exp(t,a)
    x = exp(a*t);
end;
```

```
% Sinal Exponencial
f_exp= @(t,a) exp(a*t);
```

```
>> t=linspace(-1,1,1000);
>> hold on; grid on;
>> x=f_exp(t,-2);
>> plot(t,x,'b');
>> x=f_exp(t,2);
>> plot(t,x,'r');
>> legend("a=-2","a=2");
```

```
>> t=linspace(-1,1,1000);
>> hold on; grid on;
>> x=f_exp(t,1);
>> plot(t,x,'g');
>> x=f_exp(t,2);
>> plot(t,x,'r');
>> x=f_exp(t,3);
>> plot(t,x,'b');
>> legend("a=1","a=2","a=3");
```



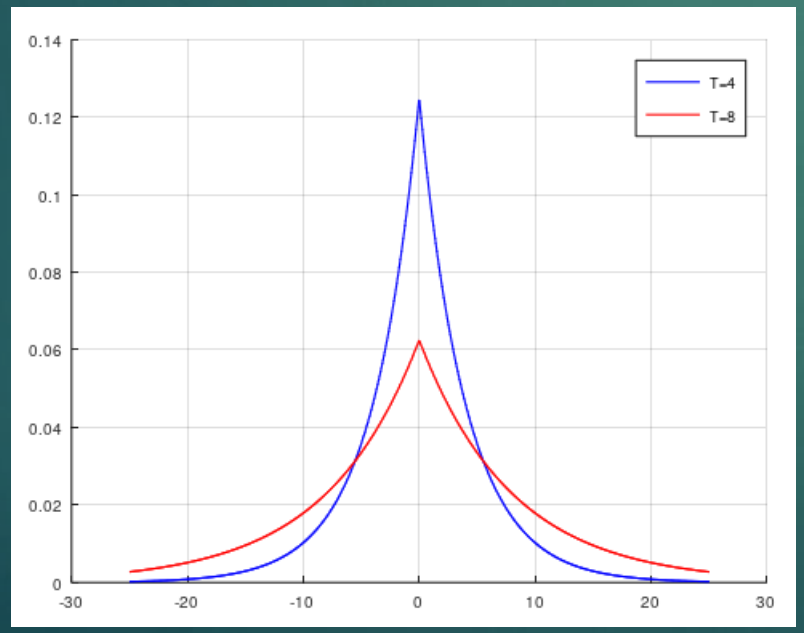


13) Pulso Exponencial

```
% Pulso Exponencial
function  x=f_pexp(t,T)
    x =1/(2*T)*exp(-abs(t)/T);
end;
```

```
% Pulso Exponencial
f_pexp= @(t,T) 1/(2*T)*exp(-abs(t)/T);
```

Contínuo : $p_T(t) = \frac{1}{2T} e^{\frac{-|t|}{T}}$ Discreto: $p_T[n] = \frac{1}{2T} e^{\frac{-|n|}{T}}$



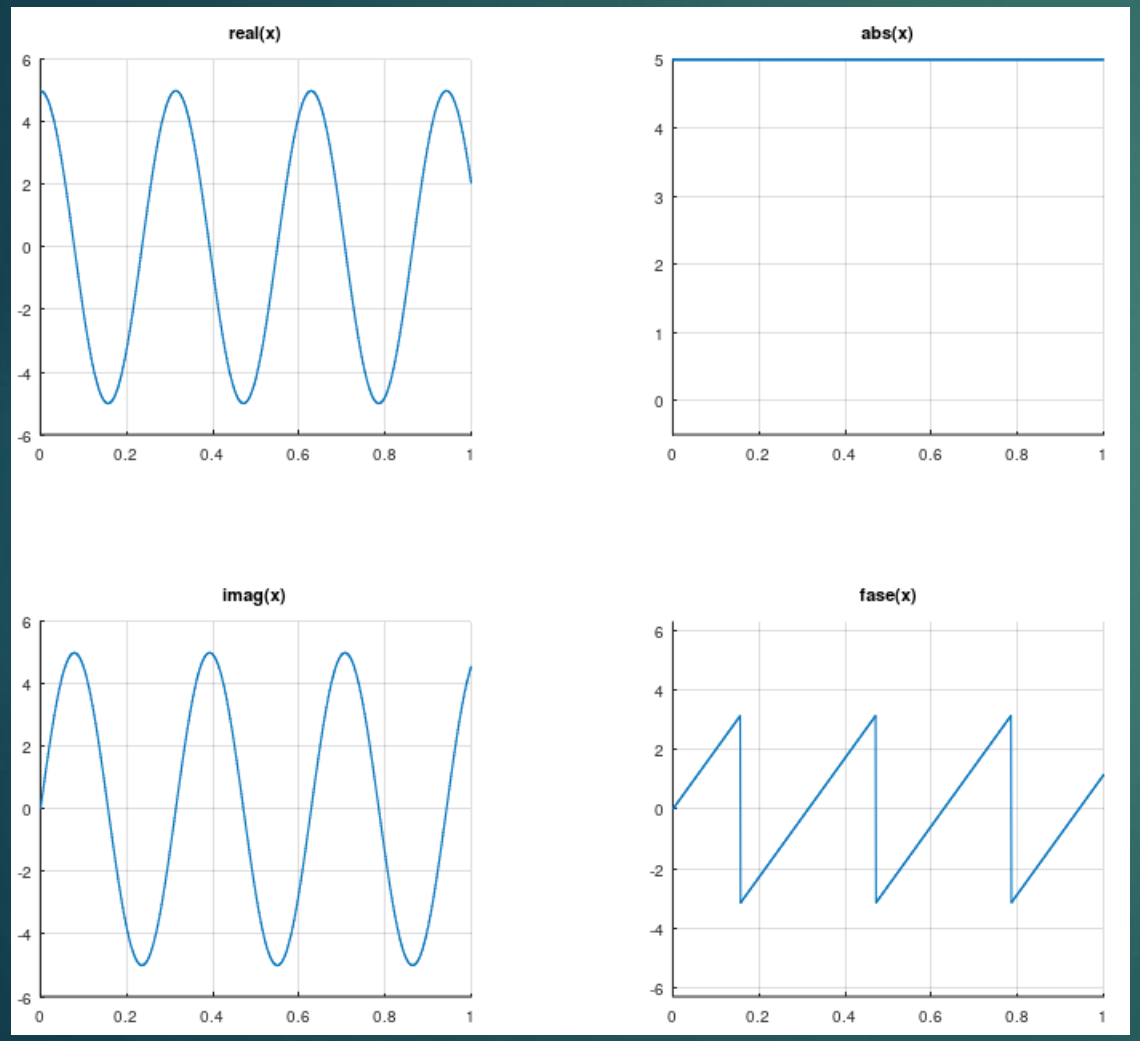

```
>> t=linspace(-25,25,1000);
>> hold on; grid on;
>> x=f_pexp(t,4);
>> plot(t,x,'b');
>> x=f_pexp(t,8);
>> plot(t,x,'r');
>> grid on;
>> legend("T=4","T=8");
```

14) Fasor

$$x(t) = e^{jwt} = \cos(w) + j\,\mathrm{sen}(w)$$

$x(t) = 5\, e^{j20t}$; 0<t<1

$w = 20$

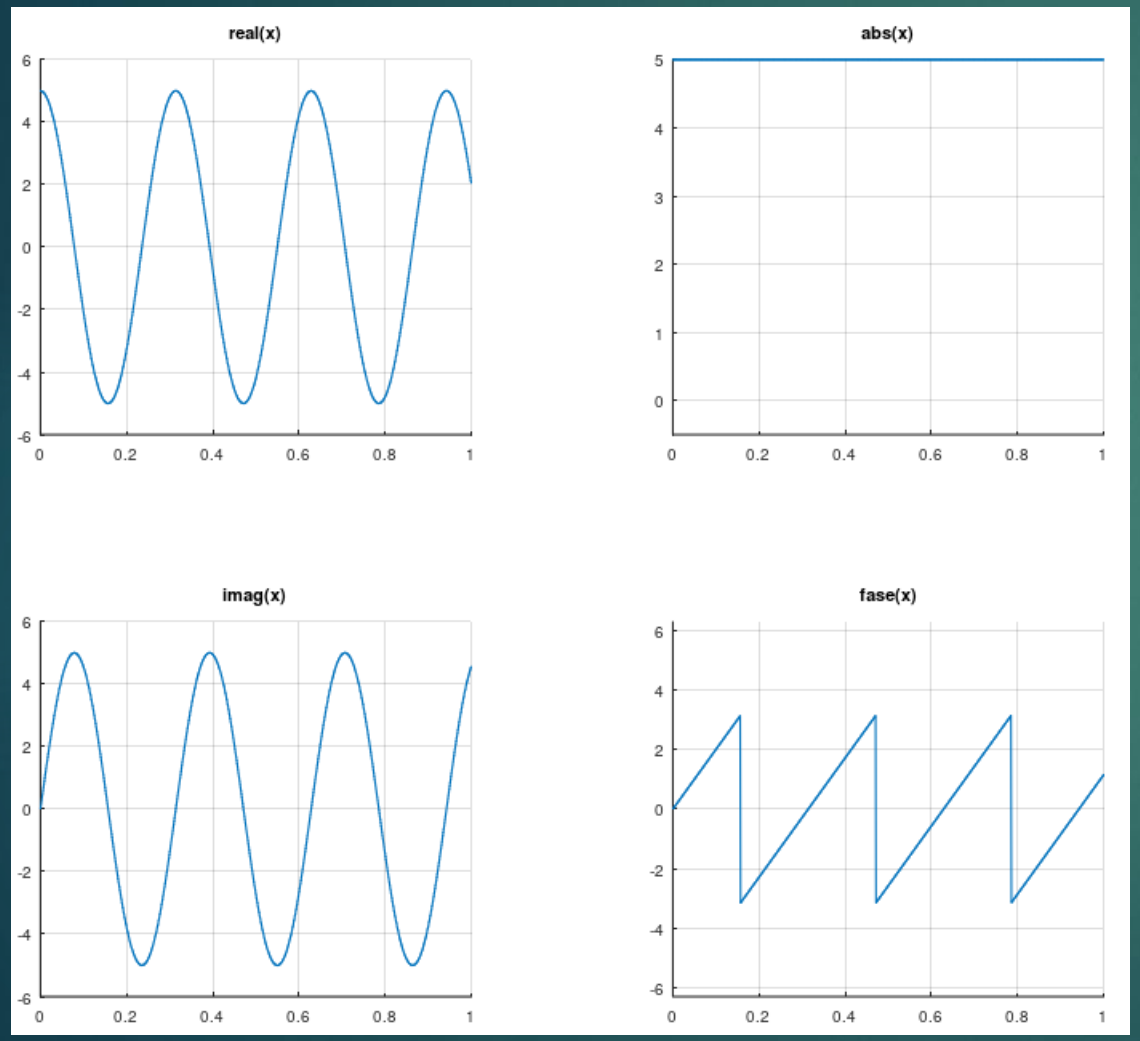


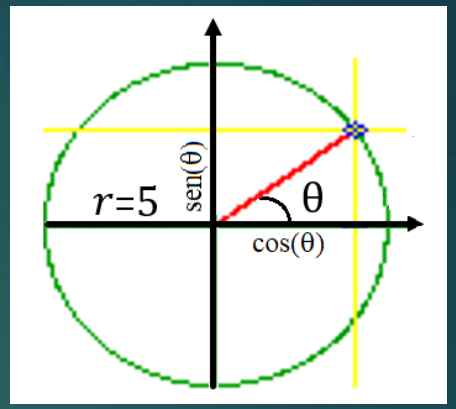

```
function plot_sinal_complexo(f,t)
    subplot(221)
      hold on; grid on;
      plot(t,real(f(t)));
      title('real(x)')
    subplot(223)
      hold on; grid on;
      plot(t,imag(f(t)));
      title('imag(x)')
    subplot(222)
      hold on; grid on;
      plot(t,abs(f(t)));
      title('abs(x)')
      lim=max(abs(f(t)));
      axis( [min(t),max(t),-0.1*lim,1.1*lim] )
    subplot(224)
      hold on; grid on;
      plot(t,atan2(imag(f(t)),real(f(t))));
      title('fase(x)')
      axis ([min(t),max(t),-2*pi,2*pi])
endfunction
```

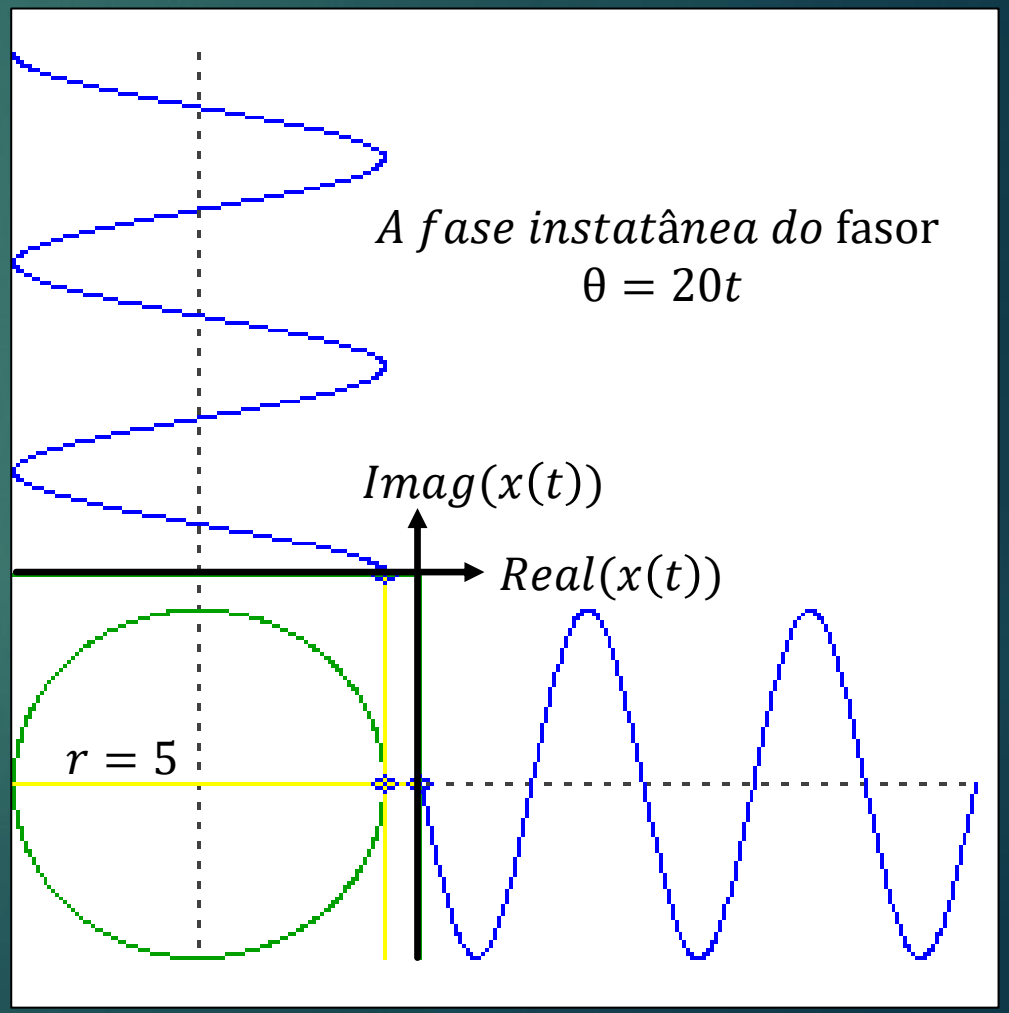
```
>> x_c = @(t) 5*exp(i*20*t);
>> t=linspace(0,1,1000);
>> plot_sinal_complexo(x_c,t)
```

Fasor em rotação

$$x(t) = 5\, e^{j20t} = 5(\cos(20t) + j\,\mathrm{sen}(20t))\ ; \quad 0<t<1$$
$$w = 20$$

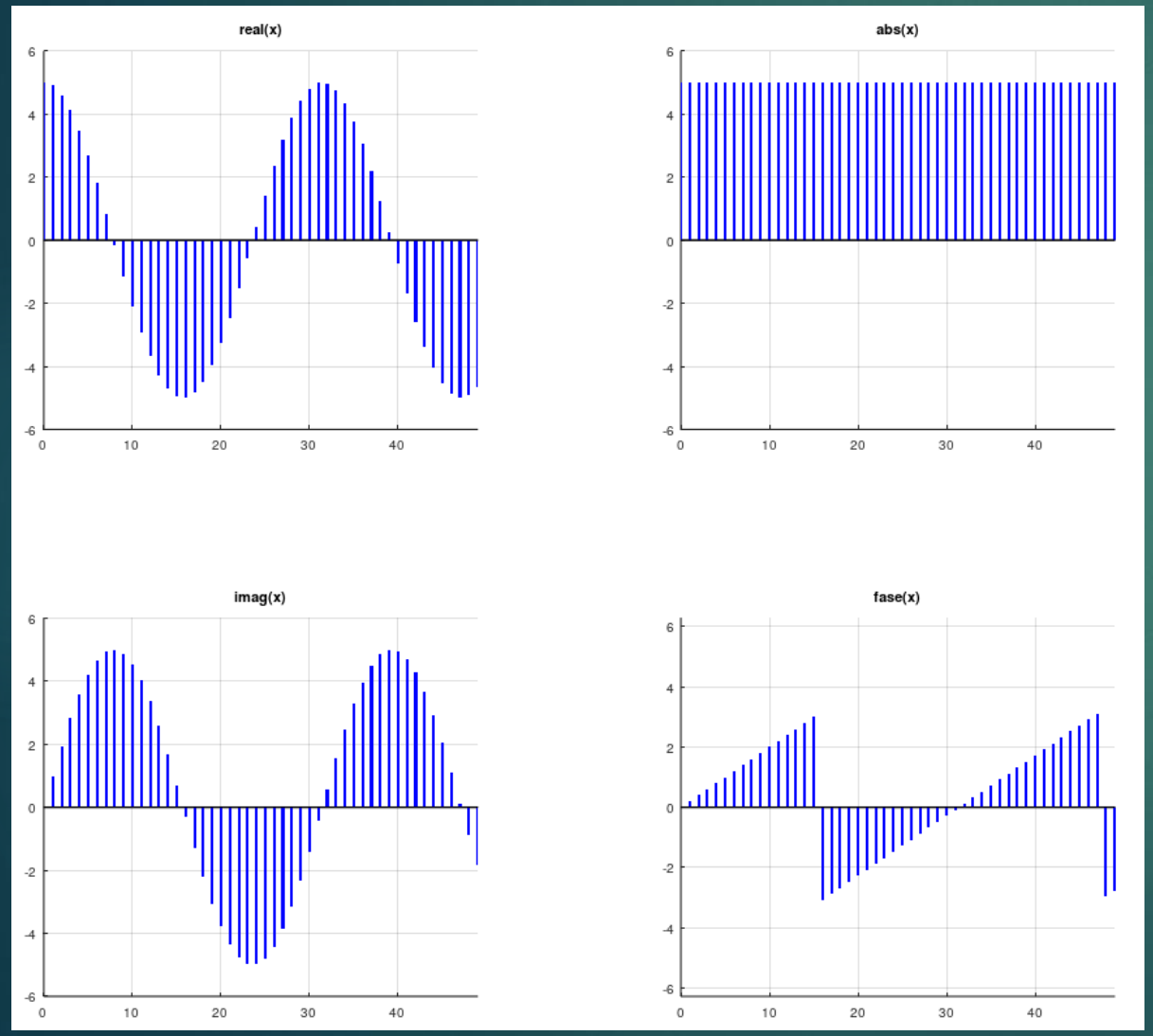






Fasor Disceto $x[n] = ke^{jw\frac{n}{N}}$

$x[n] = 5\, e^{jw\frac{n}{N}}$; 0<n<N-1

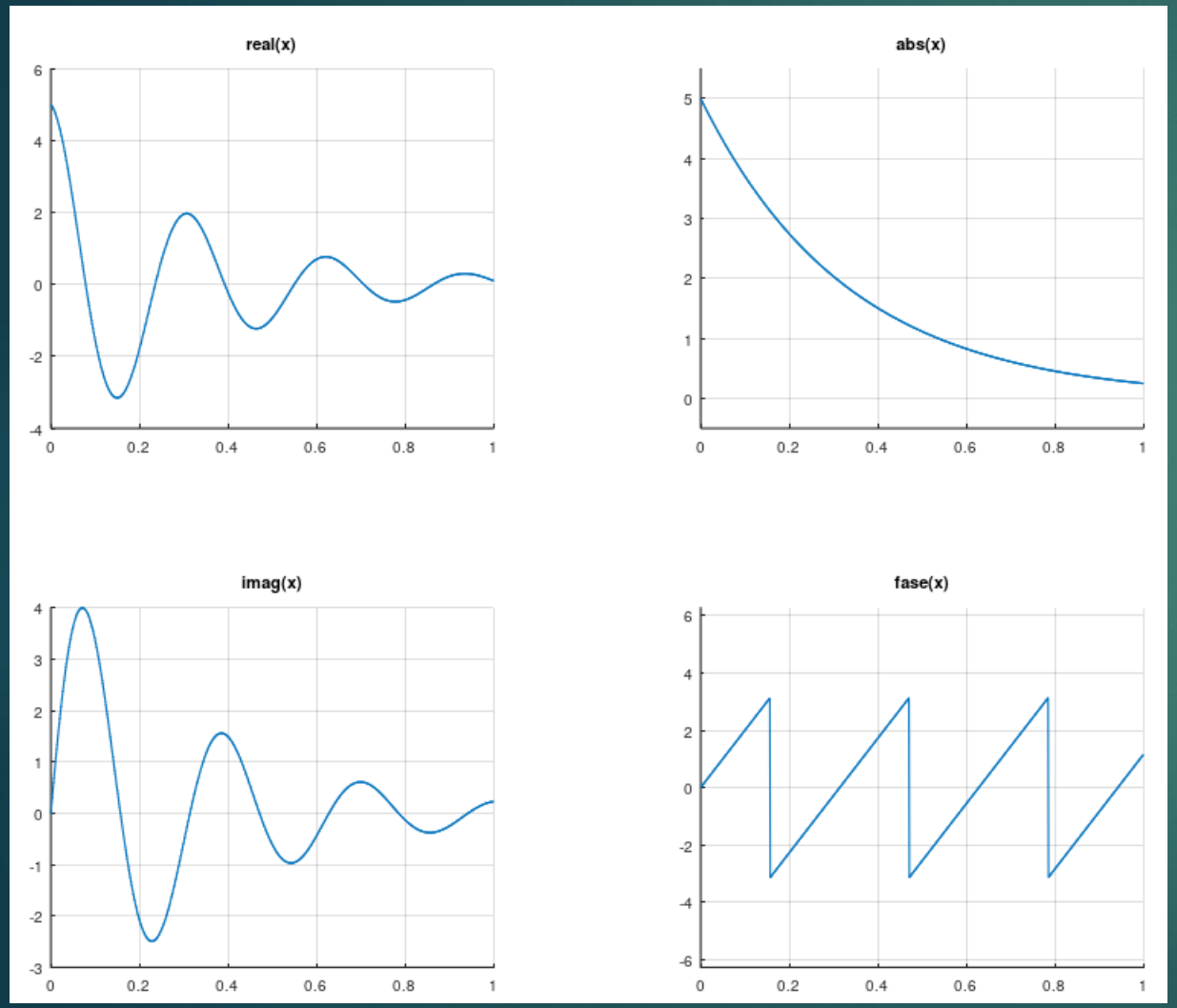



```
function plot_sinal_complexo_d(f,t)
    subplot(221)
      hold on; grid on;
      bar(t,real(f(t)),0.3,'b','edgecolor','none');
      title('real(x)')
      axis([min(t),max(t)]);
    subplot(223)
      hold on; grid on;
      bar(t,imag(f(t)),0.3,'b','edgecolor','none');
      title('imag(x)');
      axis([min(t),max(t)]);
    subplot(222)
      hold on; grid on;
      bar(t,abs(f(t)),0.3,'b','edgecolor','none');
      title('abs(x)')
      lim=max(abs(f(t)))*1.2;
      axis( [min(t),max(t),-lim,lim] )
    subplot(224)
      hold on; grid on;
      bar(t,atan2(imag(f(t)),real(f(t))),0.3,'b','edgecolor','none');
      title('fase(x)')
      axis ([min(t),max(t),-2*pi,2*pi])
endfunction
```

```
>> x_n = @(n) 5*exp((i*10)*n/50);
>> n=[0:49];
>> plot_sinal_complexo_d(x_n,n)
```

15) Sinal Exponencial Complexo

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 5*exp((-3+i*20)*t);
>> plot_sinal_complexo(x_c,t)
```



Exponencial Contínua $x(t) = e^{st}$

$s = \sigma + jw$

$x(t) = e^{st} = e^{(\sigma+jw)t}$

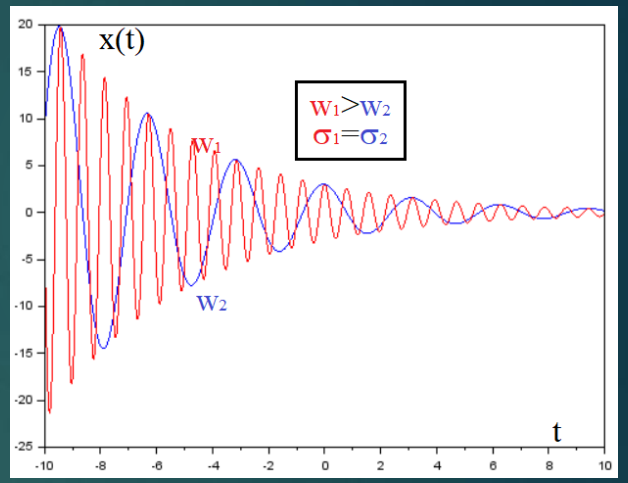
$x(t) = e^{\sigma t}e^{jwt} = e^{\sigma t}(coswt + jsenwt)$

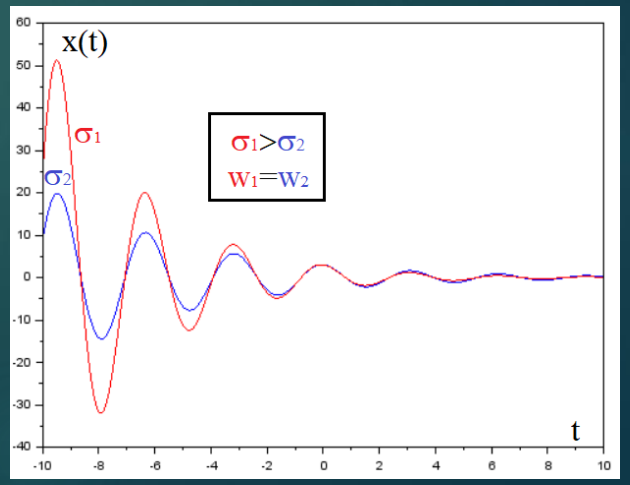
Então este sinal oscila com uma frequência w, e decai (ou cresce) exponencialmente a uma taxa σ

$x(t) = 5\, e^{(-3+j20)t}$; 0<t<1

$\alpha = -3$

$w = 20$



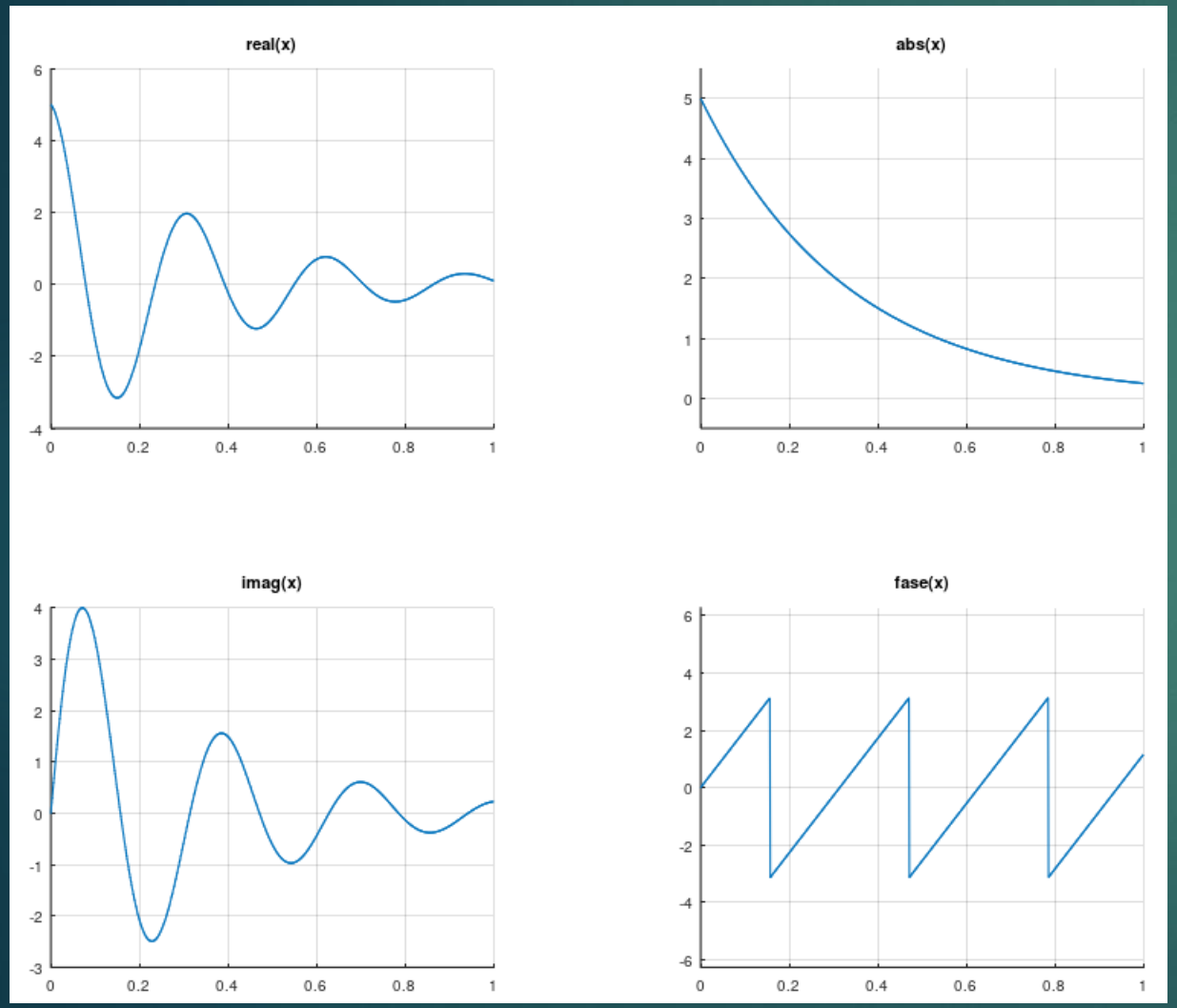


$x(t) = 5\, e^{(-3+j20)t}$; 0<t<1

$\alpha = -3$

$w = 20$

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 5*exp((-3+i*20)*t);
>> plot_sinal_complexo(x_c,t)
```

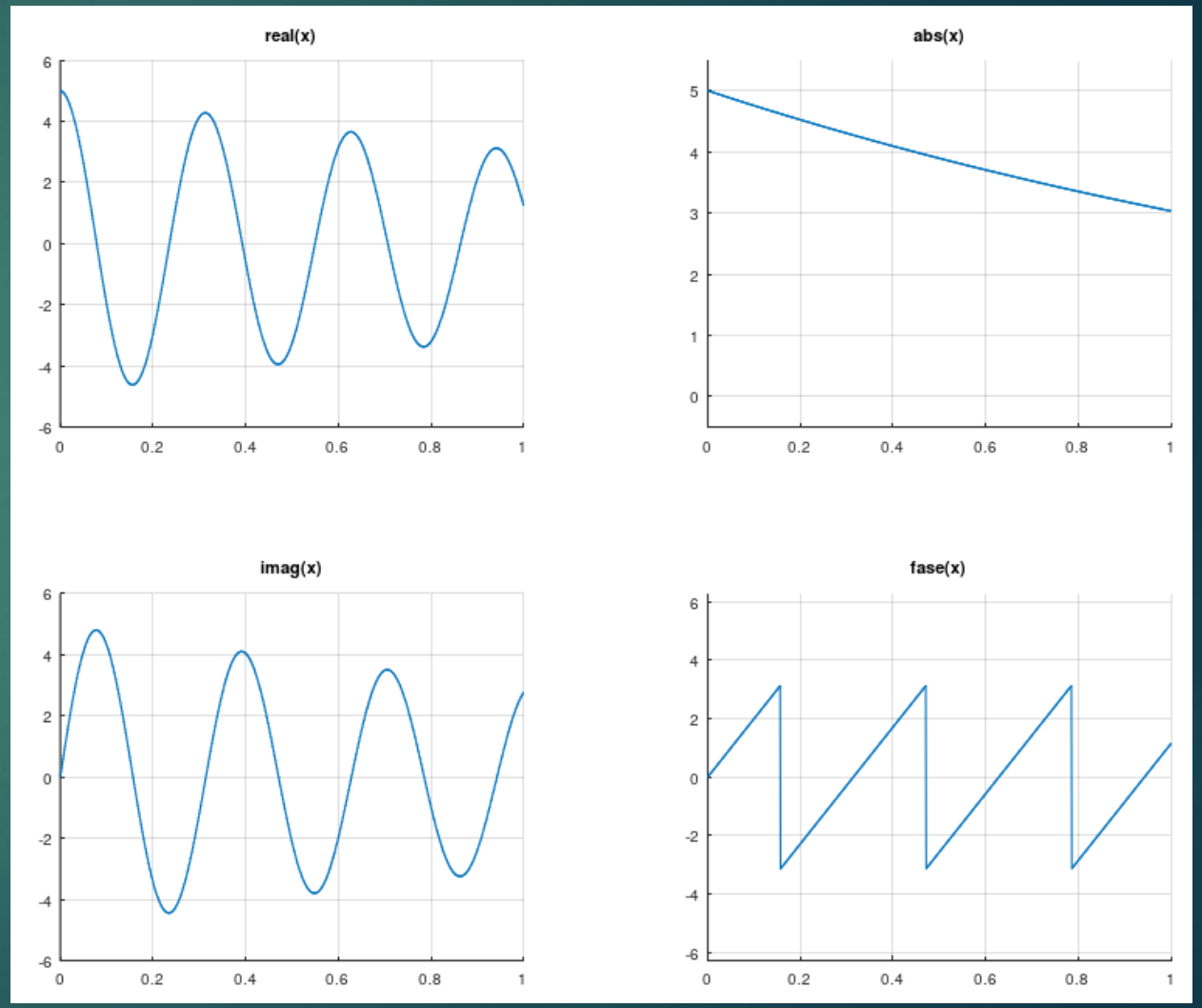


$x(t) = 5\, e^{(-0.5+j20)t}$; 0<t<1

$\alpha = -0.5$

$w = 20$

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 5*exp((-0.5+i*20)*t);
>> plot_sinal_complexo(x_c,t)
```

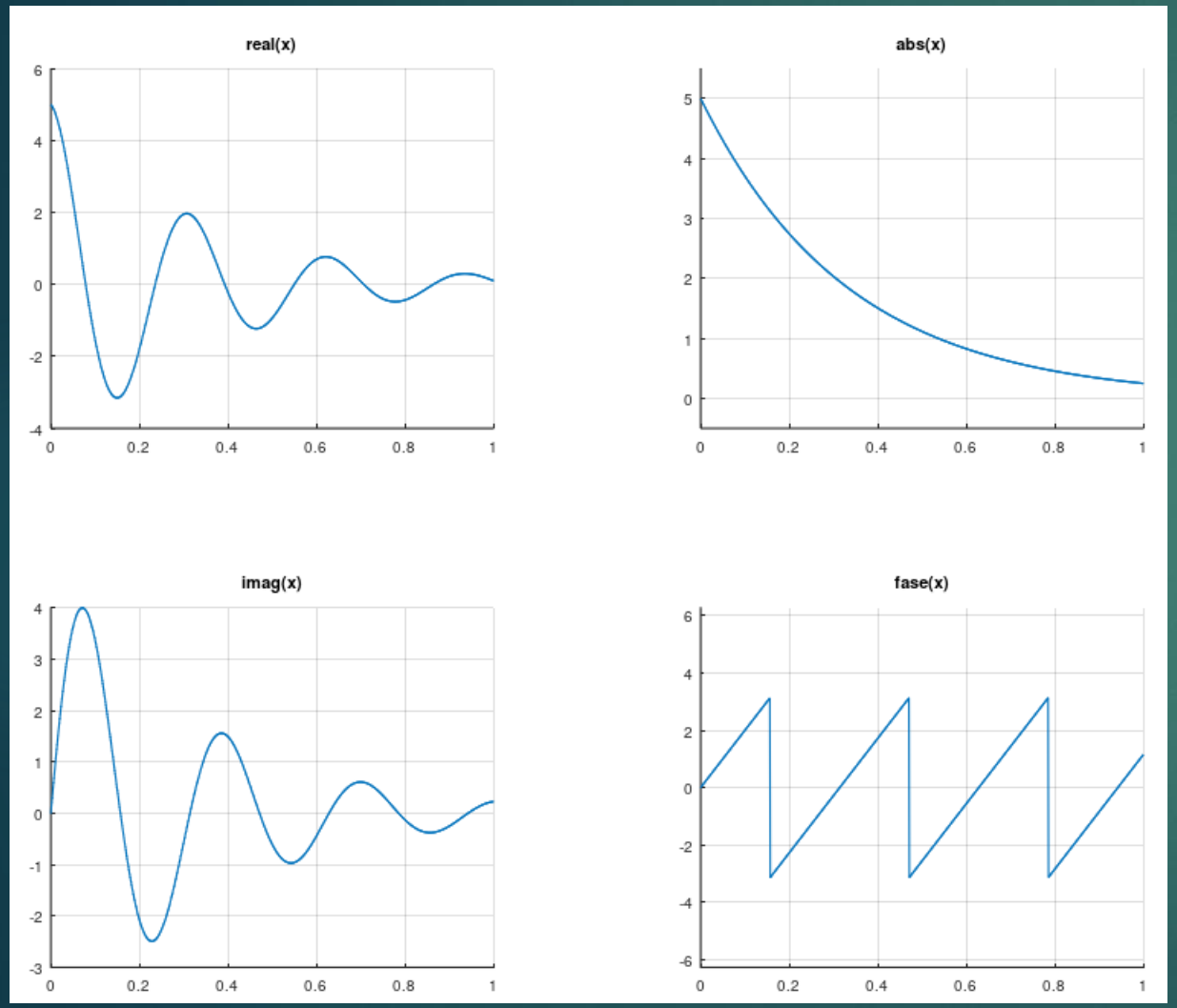


$x(t) = 5\, e^{(-3+j20)t}$; 0<t<1

$\alpha = -3$

$w = 20$

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 5*exp((-3+i*20)*t);
>> plot_sinal_complexo(x_c,t)
```

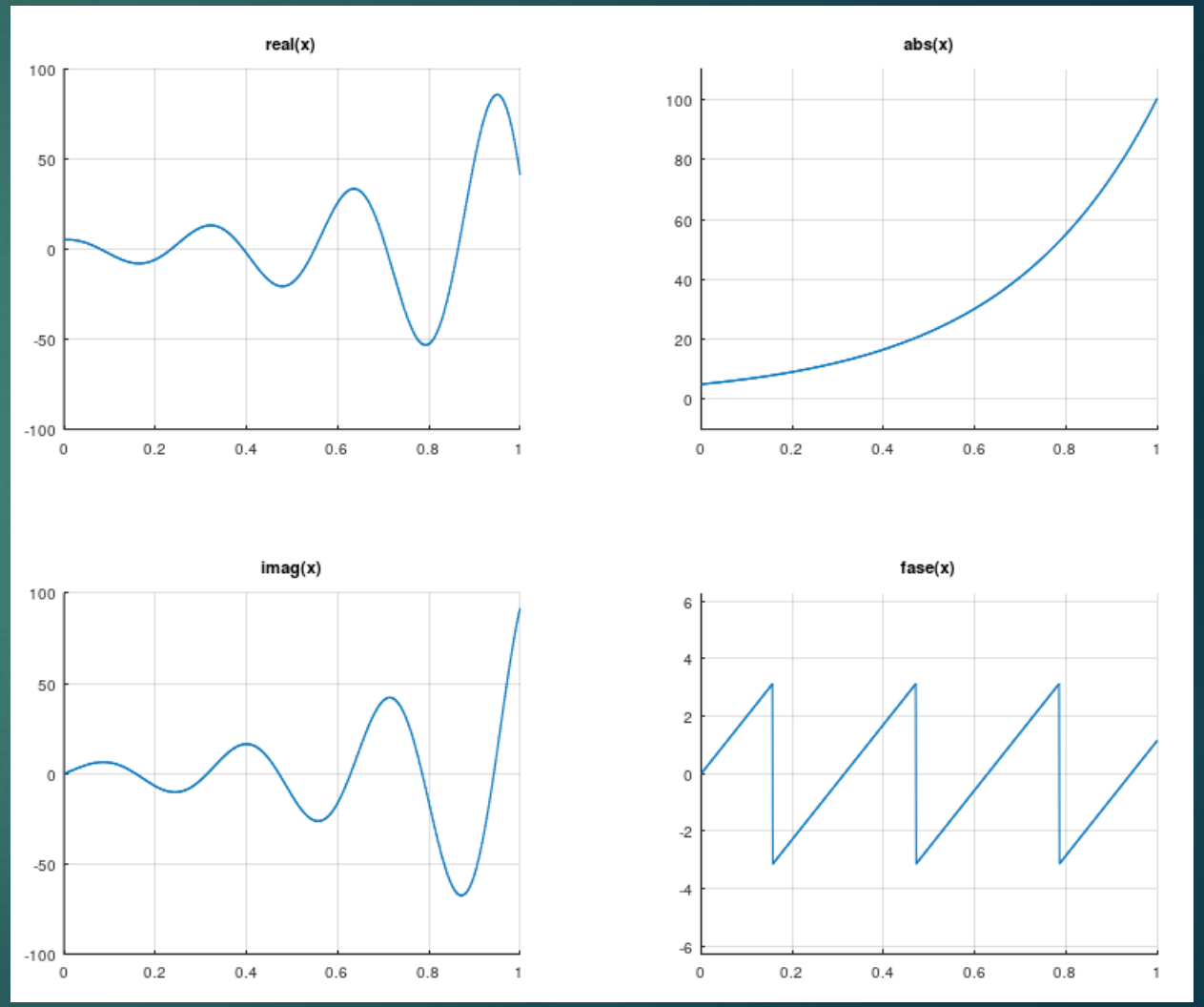


$x(t) = 5\, e^{(+3+j20)t}$; 0<t<1

$\alpha = +3$

$w = 20$

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 5*exp((3+i*20)*t);
>> plot_sinal_complexo(x_c,t)
```

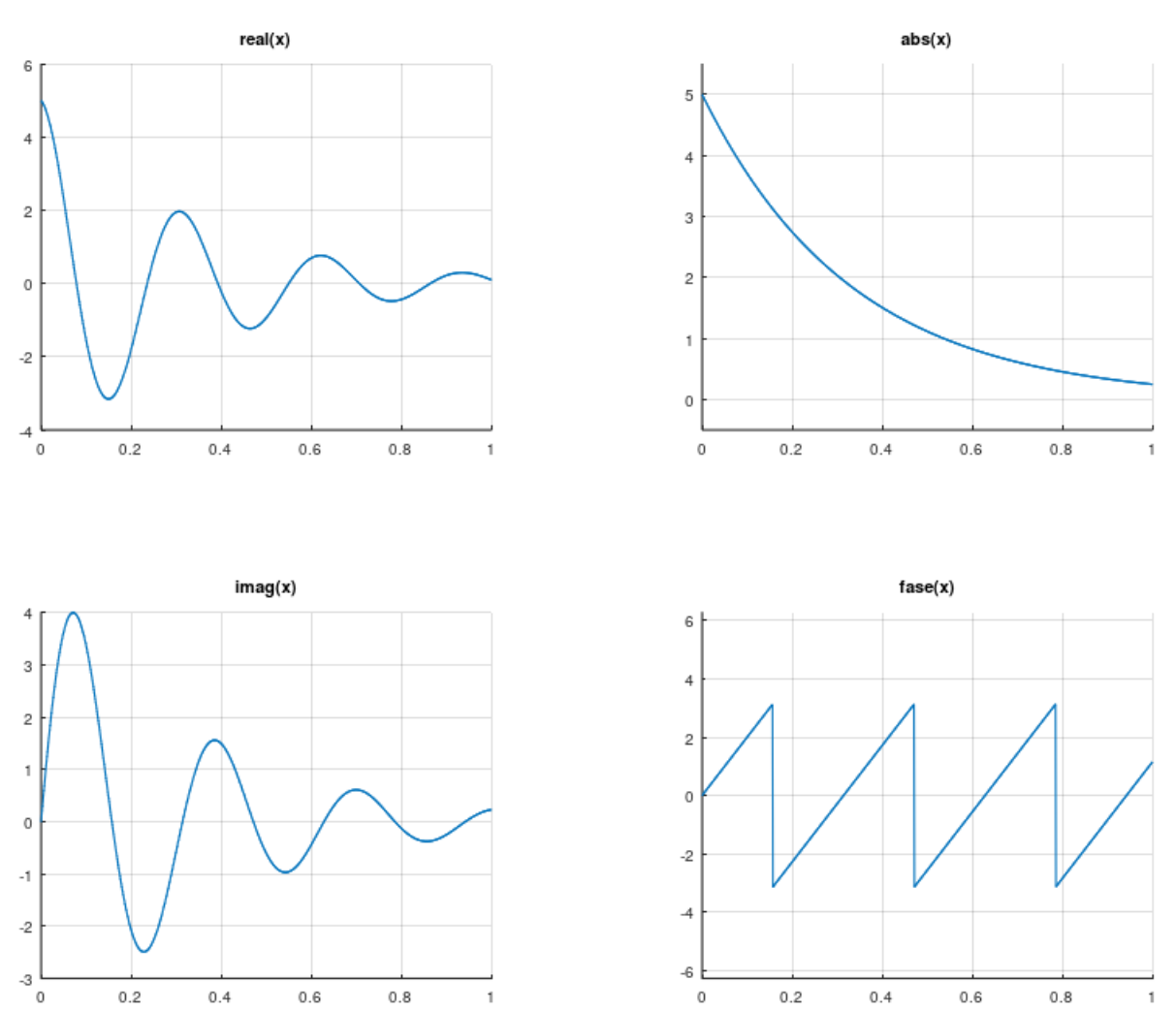


$x(t) = 5\, e^{(-3+j20)t}$; 0<t<1

$\alpha = -3$

$w = 20$

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 5*exp((-3+i*20)*t);
>> plot_sinal_complexo(x_c,t)
```

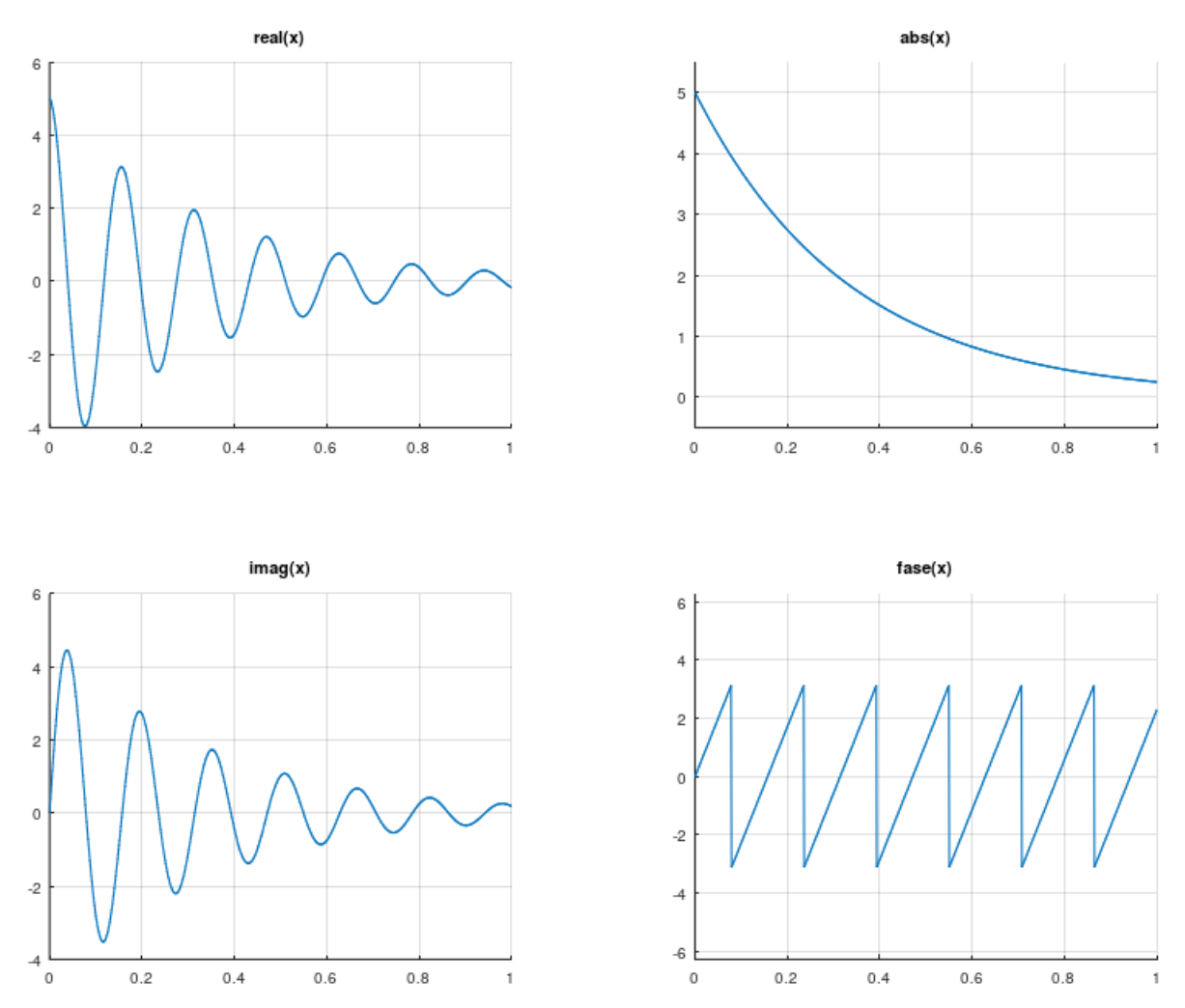


$x(t) = 5\, e^{(-3+j40)t}$; 0<t<1

$\alpha = -3$

$w = 40$

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 5*exp((-3+i*40)*t);
>> plot_sinal_complexo(x_c,t)
```

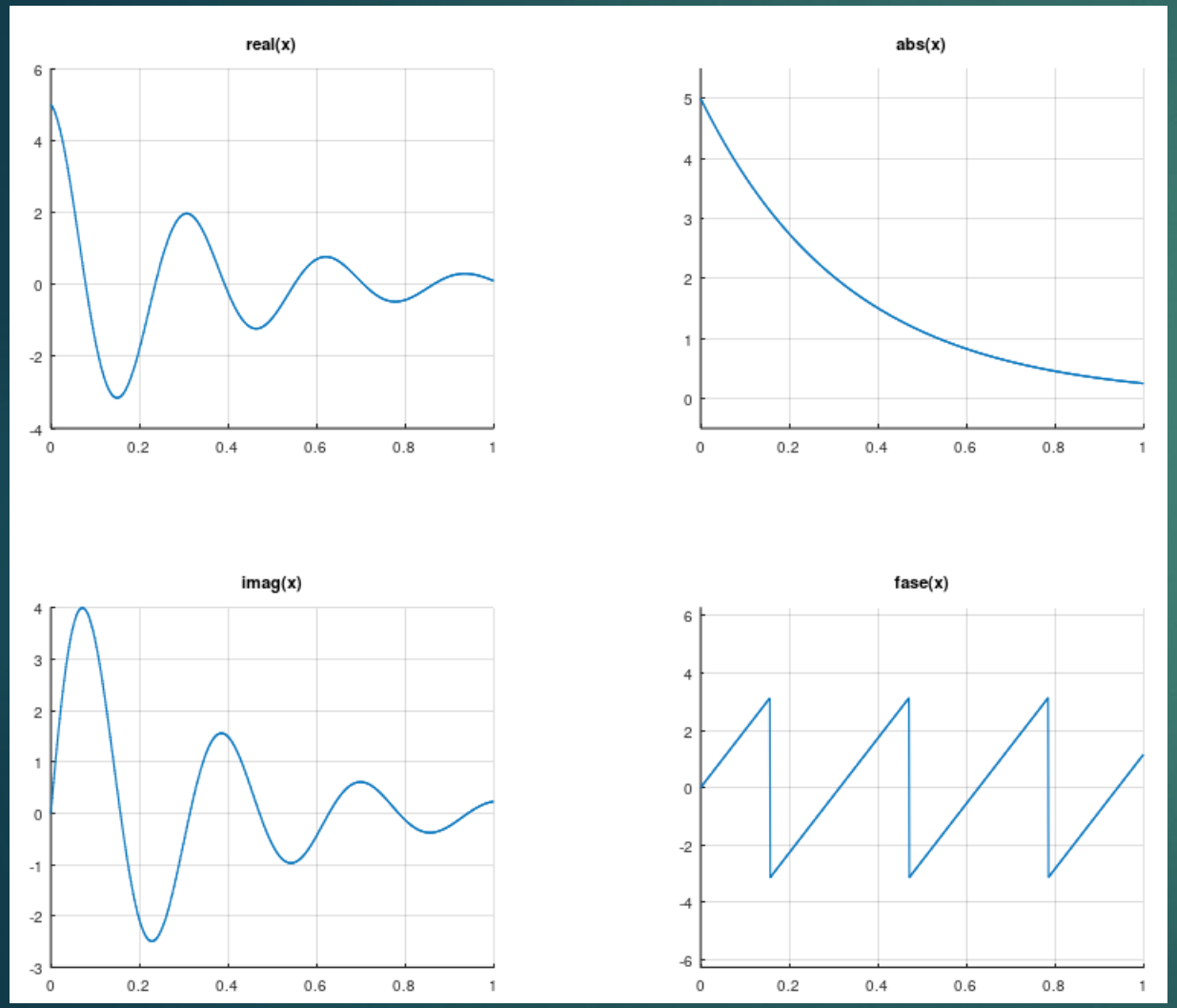


$x(t) = 5\, e^{(-3+j20)t}$; 0<t<1

$\alpha = -3$

$w = 20$

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 5*exp((-3+i*20)*t);
>> plot_sinal_complexo(x_c,t)
```

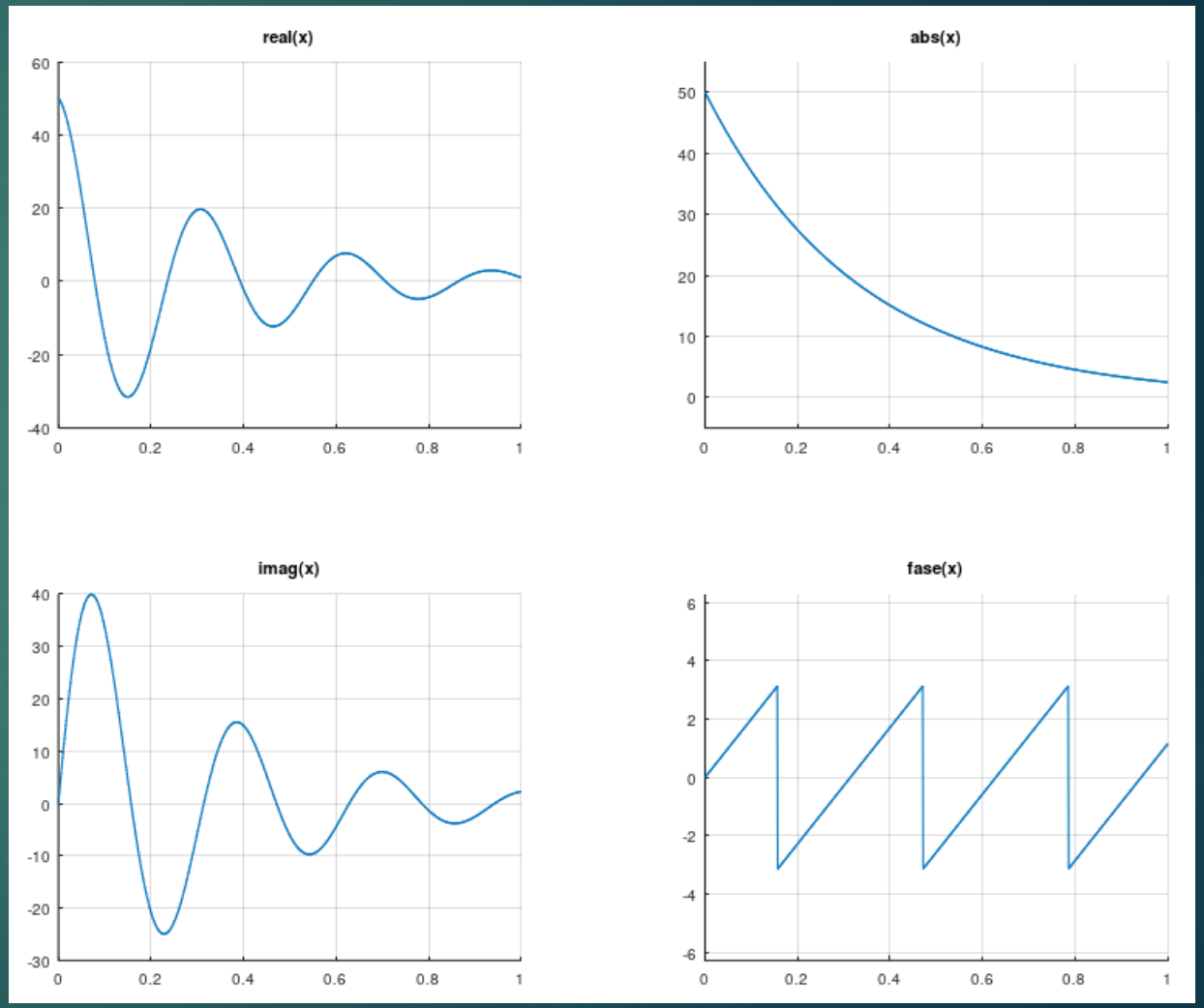


$x(t) = 50\, e^{(-3+j20)t}$; 0<t<1

$\alpha = -3$

$w = 20$

```
>> t=linspace(0,1,1000);
>> x_c = @(t) 50*exp((-3+i*20)*t);
>> plot_sinal_complexo(x_c,t)
```



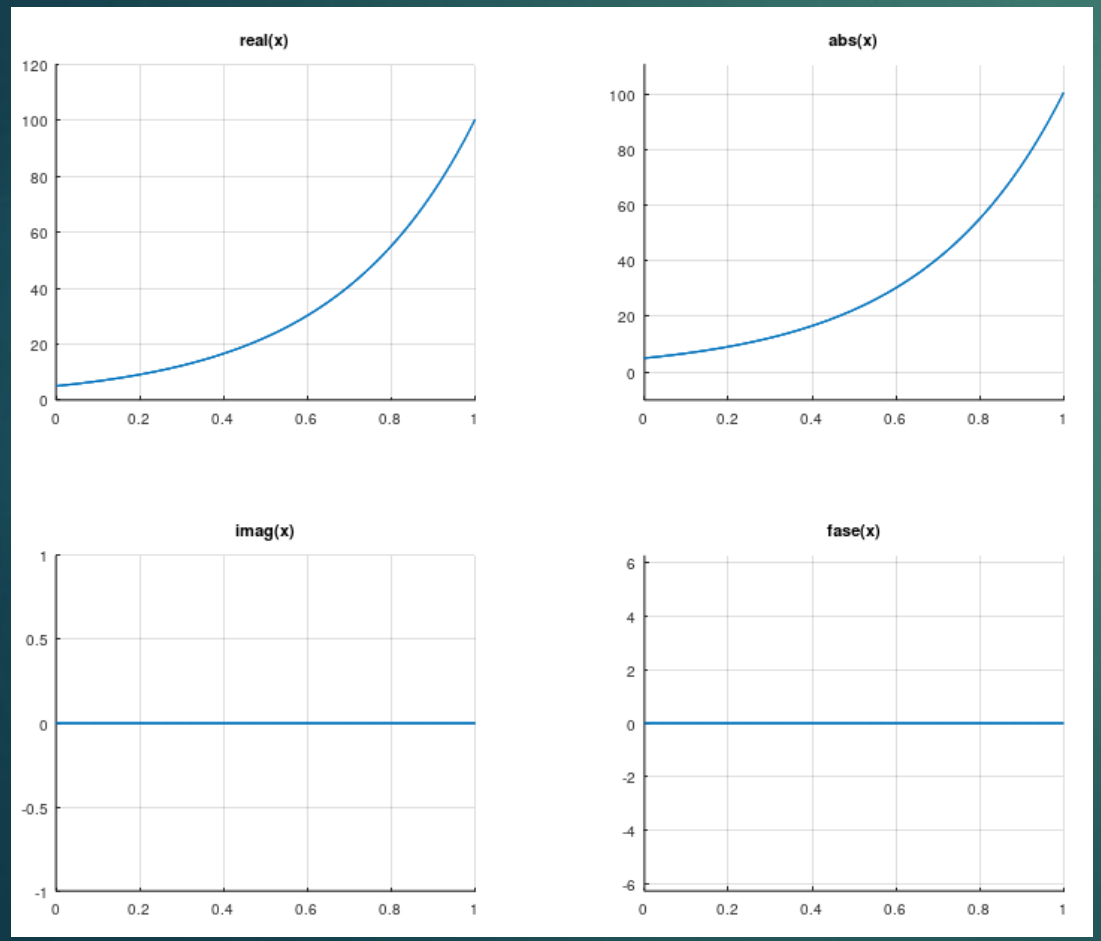
Casos especiais do Sinal Exponencial Complexo :

Sinal Exponencial Real (σ≠ 0; w=0) $x(t) = e^{\sigma t}$

- Veja Item 12

$x(t) = 5\, e^{(-3+j0)t}$;
0<t<1
$\alpha = -3$
$w = 0$

```
>> x_c = @(t) 5*exp((3+i*0)*t);
>> t=linspace(0,1,1000);
>> plot_sinal_complexo(x_c,t)
```

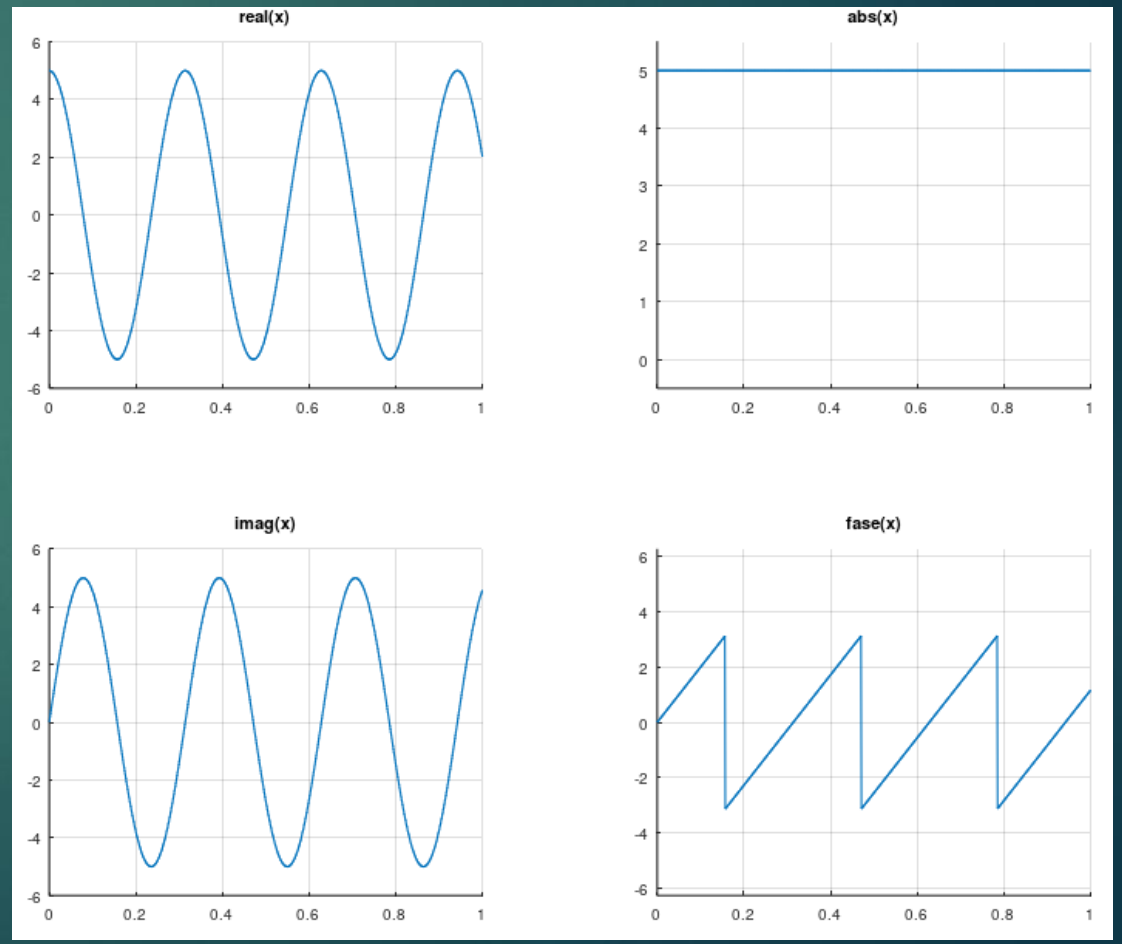


Fasor (σ=0 ; $w \neq 0$)! $x(t) = e^{jwt}$

- Veja Item 14

$x(t) = 5\, e^{(0+j20)t}$;
0<t<1
$\alpha = 0$
$w = 20$

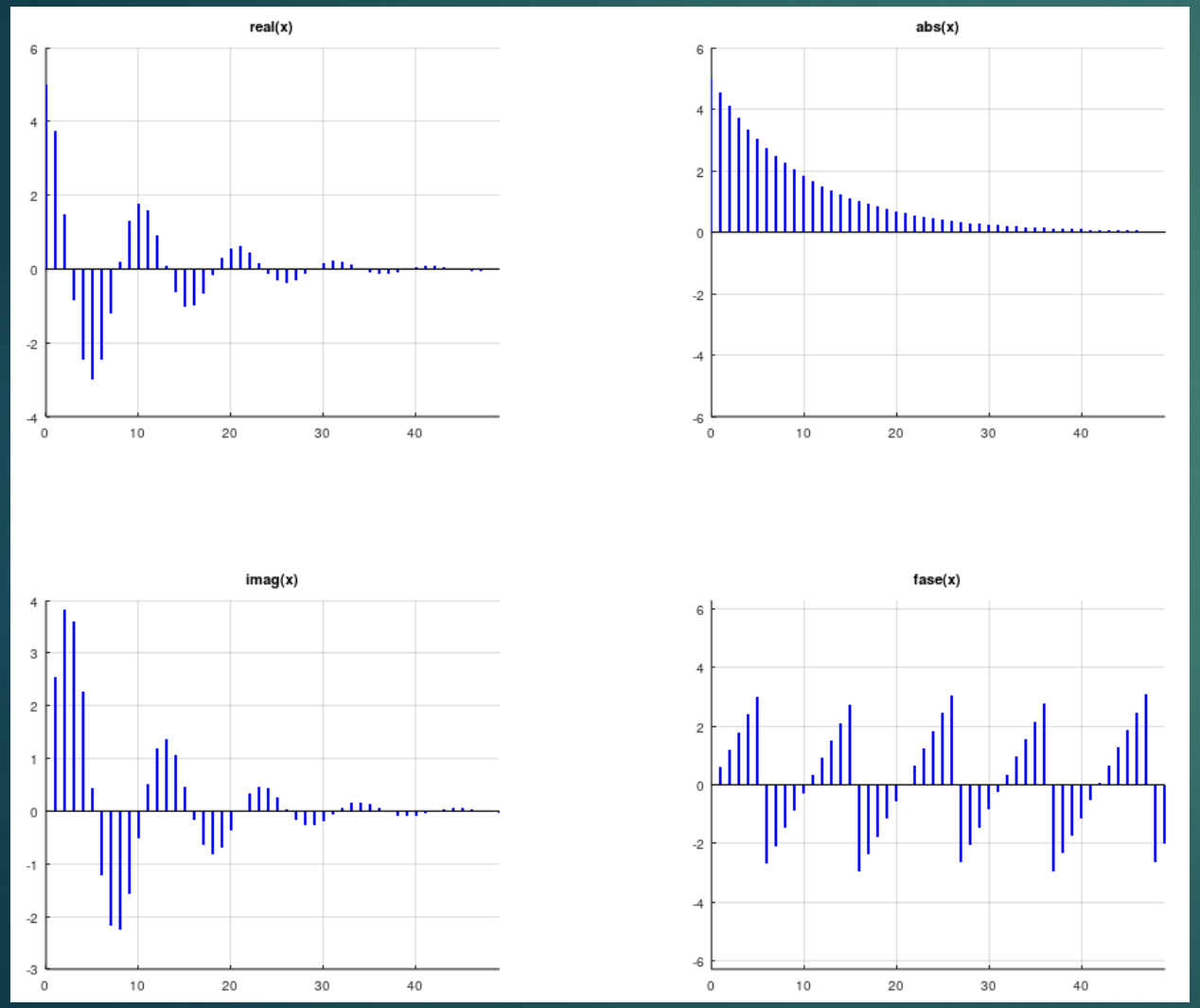
```
>> x_c = @(t) 5*exp((0+i*20)*t);
>> t=linspace(0,1,1000);
>> plot_sinal_complexo(x_c,t)
```



$$x[n] = 20\, e^{\frac{(-5+j30)n}{N}} \; ; \quad N=50; \quad 0 \le n < 50$$

$$\alpha = -5/N$$

$$w = 30/N$$




```
>> x_n = @(n) 5*exp((-5+i*30)*n/50);
>> n=[0:49];
>> plot_sinal_complexo_d(x_n,n)
>>
```

16) Senoides Contínuas(Senos e Cossenos)

$$x(t) = \cos(wt + \theta)$$

$$w = 2\pi f = \frac{2\pi}{T}$$

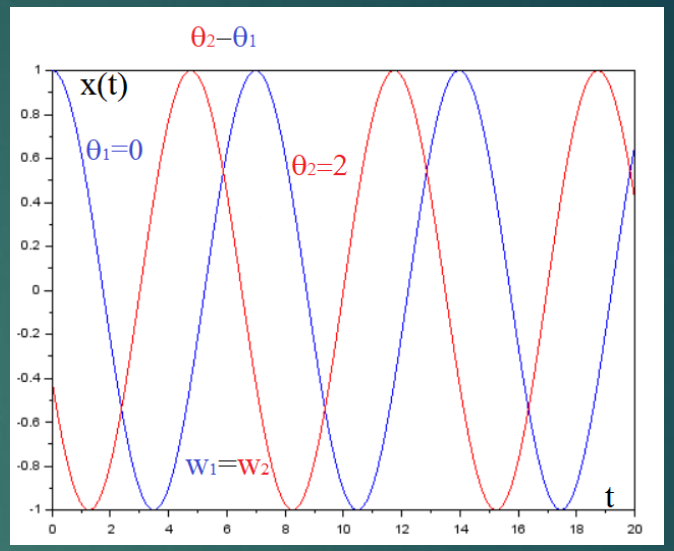
$$T = \frac{2\pi}{w} = \frac{1}{f}$$

w = frequência angular em rad/s

T = Período em segundos

f = frequência em Hertz

θ = Fase em radianos



Exemplo: $x(t) = \cos(\frac{2\pi}{3}t)$

$$w = \frac{2\pi}{3}$$

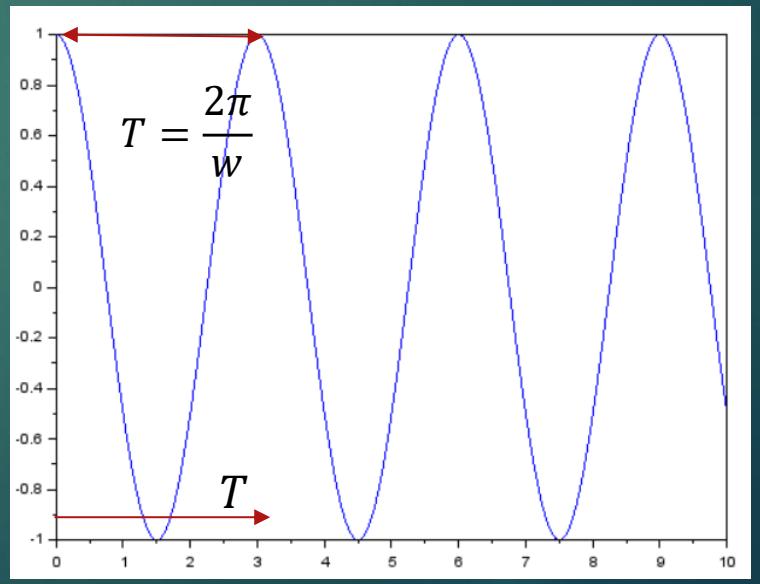
T = 3

f $= \frac{1}{3}$

$\theta = 0$

$$x(0) = \cos\left(\frac{2\pi}{3}0\right) = 1$$

$$x(T) = \cos\left(\frac{2\pi}{3}3\right) = \cos(2\pi) = 1$$



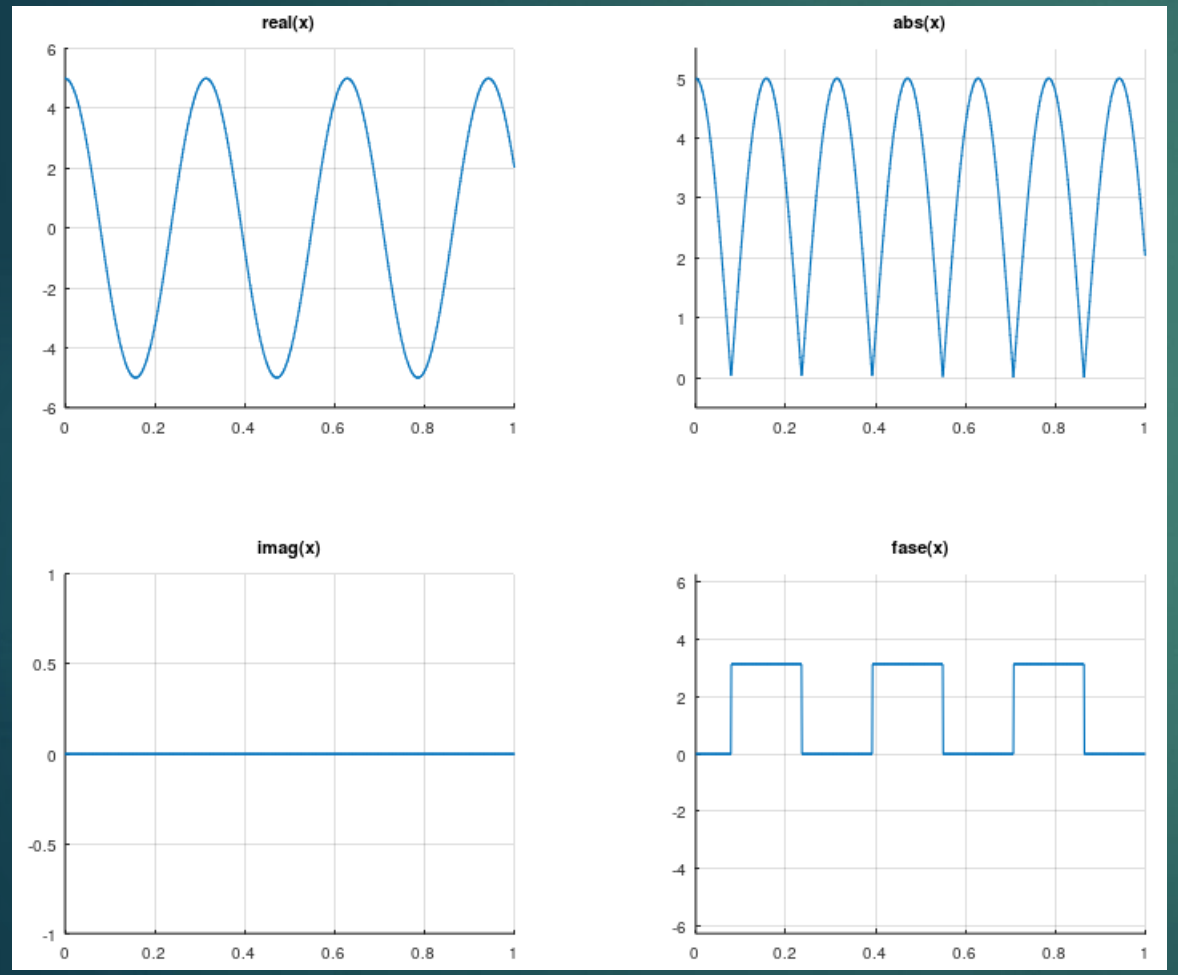
$x(t) = 5\cos(20t)$; 0<t<1

$k = 5$

$w = 20$

$\theta = 0$

```
>> x_c = @(t) 5*cos(20*t);
>> t=linspace(0,1,1000);
>> plot_sinal_complexo(x_c,t)
```



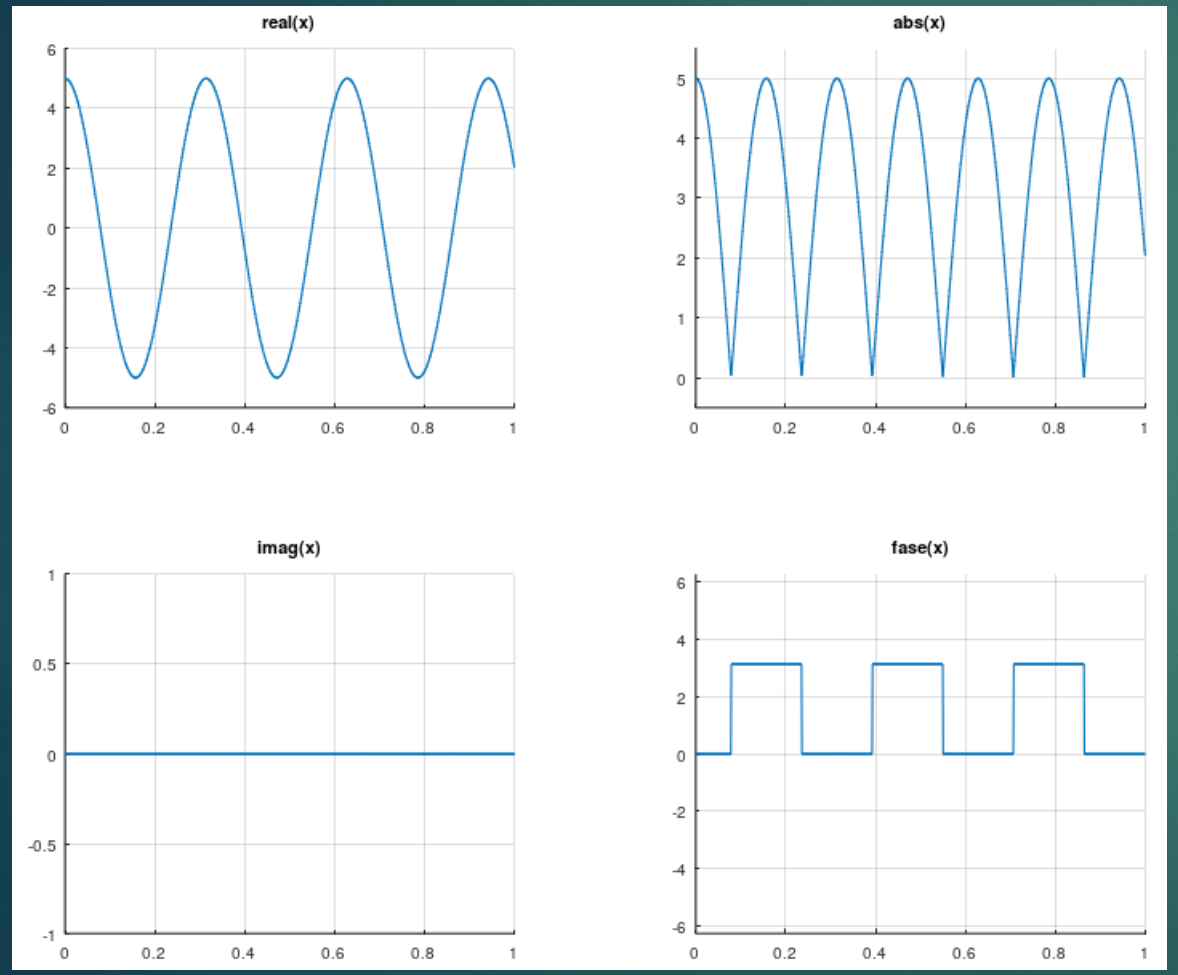
$x(t) = 5\cos(20t)$; 0<t<1

$k = 5$

$w = 20$

$\theta = 0$

```
>> x_c = @(t) 5*cos(20*t);
>> t=linspace(0,1,1000);
>> plot_sinal_complexo(x_c,t)
```



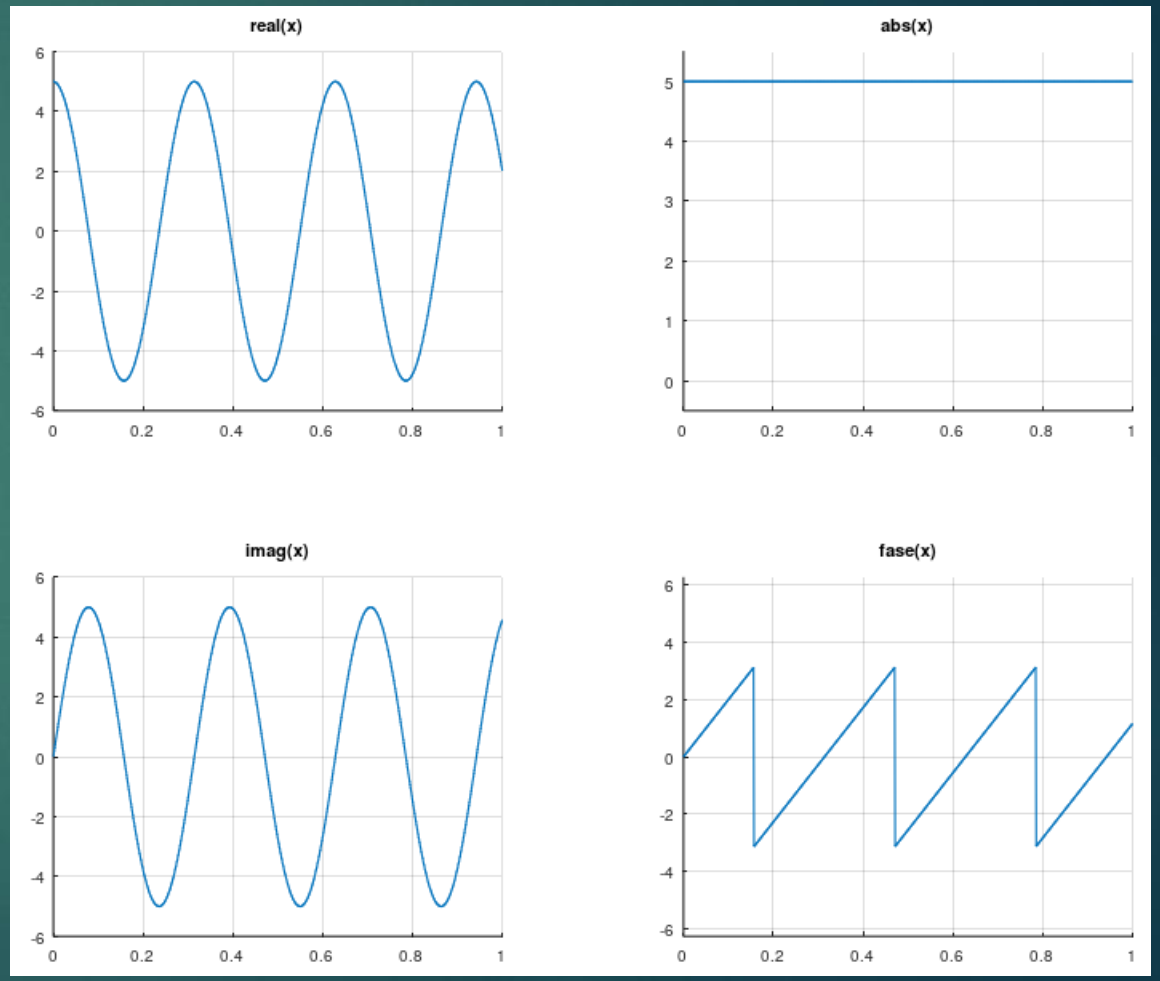
$x(t) = 5\,e^{j20t}$; 0<t<1

$k = 5$

$w = 20$

$\theta = 0$

```
>> x_c = @(t) 5*exp(i*20*t);
>> t=linspace(0,1,1000);
>> plot_sinal_complexo(x_c,t)
```



17) Função Cosseno/Seno Discreto

$$x[n] = k\ \cos(w_d\, n + \theta)$$

$w_d =$ frequência angular em rad/amostra

Exemplo:

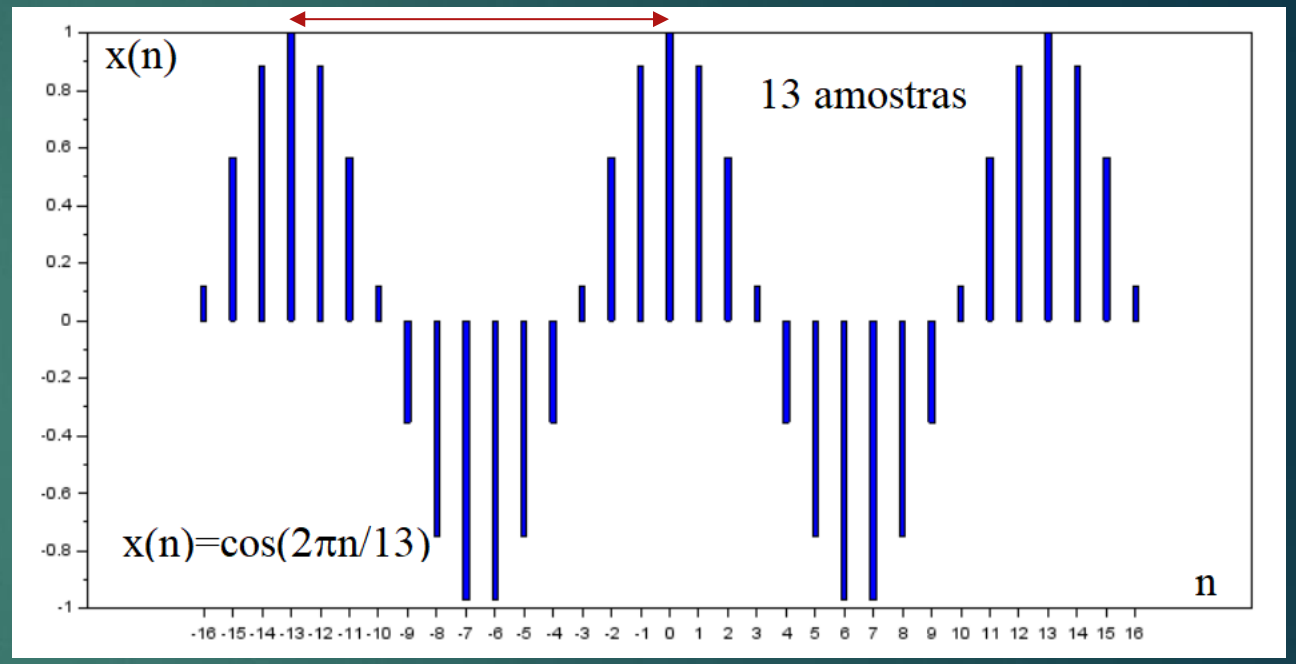
$$x[n] = \cos\left(\frac{2\pi}{13} n\right)$$

$$w_d = \frac{2\pi}{13}$$

$$x[0] = \cos\left(\frac{2\pi}{13} 0\right) = 1$$

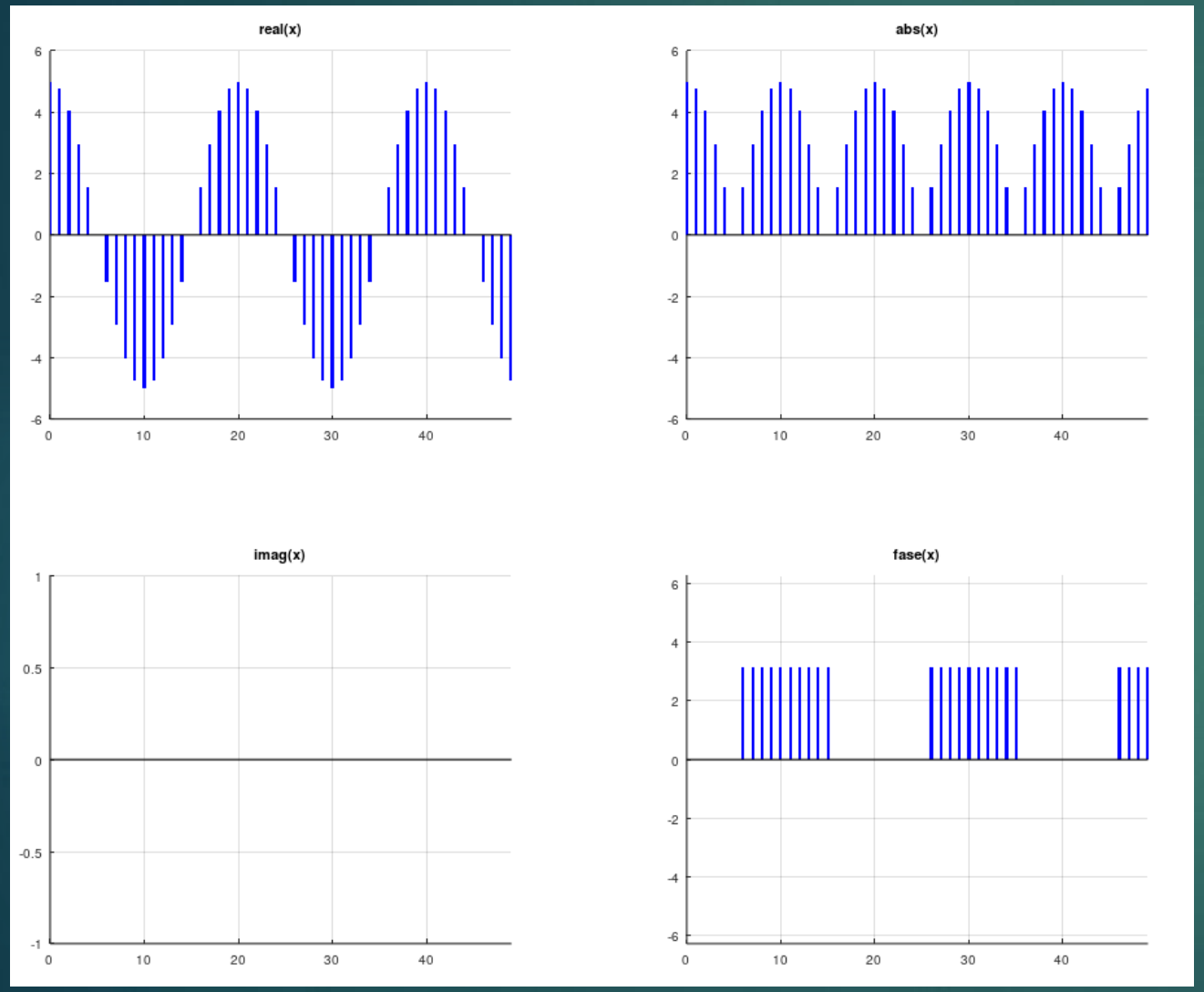
$$x[13] = \cos\left(\frac{2\pi}{13} 13\right) = 1$$

$$N = \frac{2\pi}{\mathrm{w_d}} = 13$$



Isso significa que em um ciclo completo (2π), teremos 13 amostras!

Função $x[n] = k\,cos(wn + \theta)$



$x[n] = 5\,cos\left(\frac{5\pi}{N}n\right)$; 0<n<N-1;

N=50

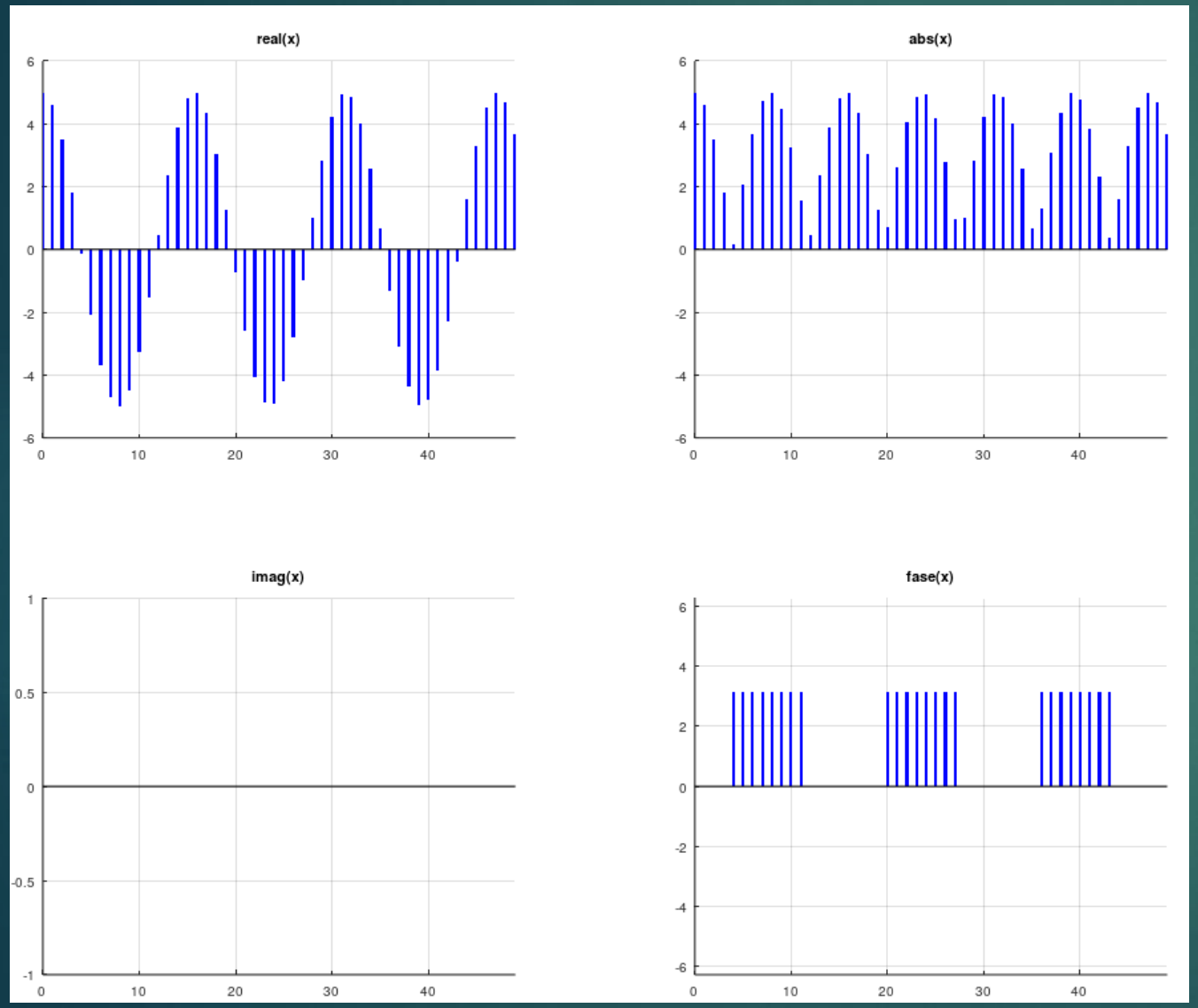
$\theta = 0$

$$w_d = \frac{5\pi}{50} = \frac{2\pi}{20}$$

(ciclo completo em 20 *amostras*)

```
>> x_n = @(n) 5*cos((5*pi)*n/50);
>> n=[0:49];
>> plot_sinal_complexo_d(x_n,n)
```

Função $x[n] = k\,cos(wn + \theta)$



$x[n] = 5\,cos\left(\frac{20}{N}n\right)$; 0<n<N-1;

N=50

$\theta = 0$

$w_d = \frac{20}{50} = \frac{2\pi}{5\pi}$

(ciclo completo em $5\pi = 15.707963 \ldots$ *amostras*?!?!)

```
>> x_n = @(n) 5*cos((20)*n/50);
>> n=[0:49];
>> plot_sinal_complexo_d(x_n,n)
>>
```

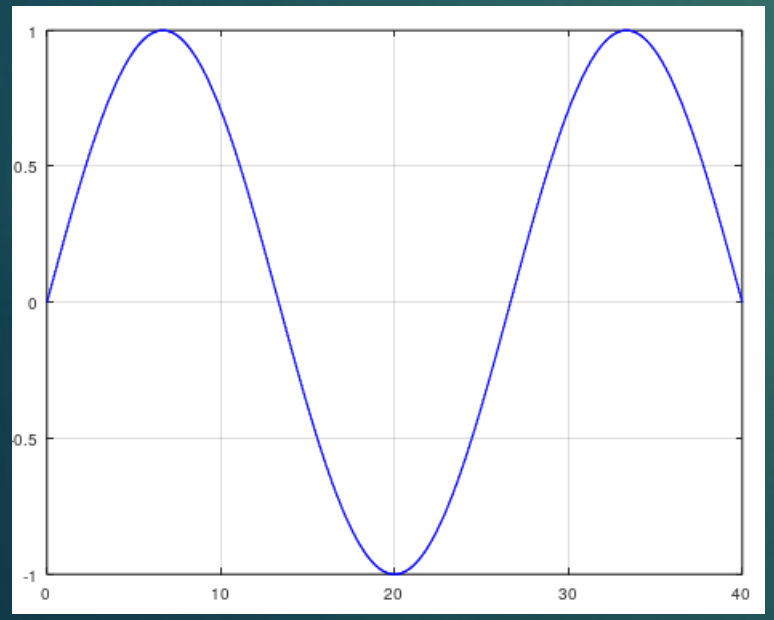
Classes de sinais

1) SINAIS CONTÍNUOS
2) SINAIS DISCRETOS
3) SINAIS PARES
4) SINAIS ÍMPARES
5) SINAIS PERIÓDICOS
6) SINAIS APERIÓDICOS
7) SINAIS DETERMINÍSTICOS
8) SINAIS ALEATÓRIOS

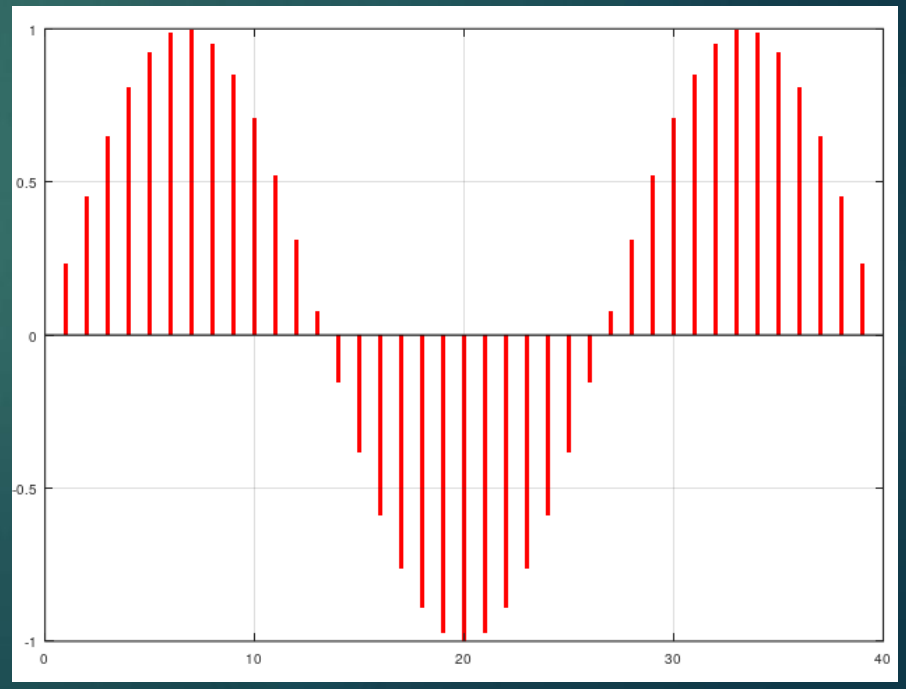
Sinais contínuos X sinais discretos

- Nos sinais de tempo discreto, a variável independente e contínua 't' é amostrada em instantes discretos de tempo (índice 'n'). Então estes sinais não são representados por uma curva contínua $x(t)$, mas por uma sequência de números $x[n]$ (o que é mais adequado para processamento em computador).

```
>> x_c = @(t) sin(2*pi*(1.5/40)*t);
>> t=linspace(0,40,1001);
>> plot(t,x_c(t),'b');
>> grid on;
```

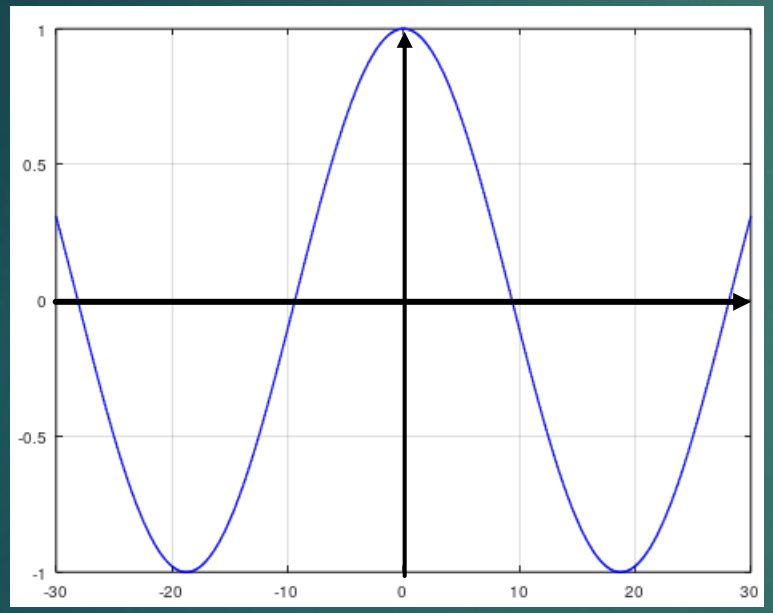



```
>> x_c = @(t) sin(2*pi*(1.5/40)*t);
>> n=[0:40];
>> bar(n,x_c(n),0.2,'r','edgecolor','none');
>> grid on;
>> axis([min(n),max(n)])
```

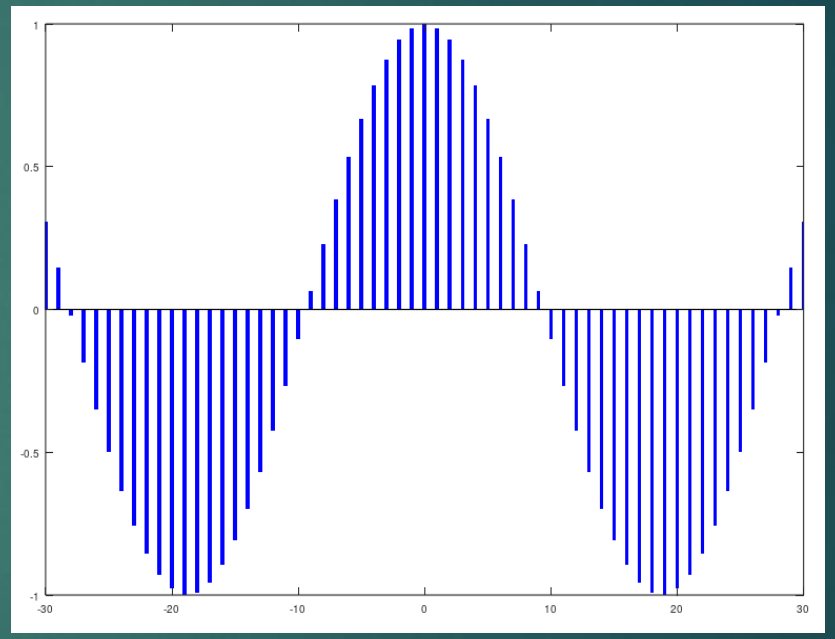


Sinais Pares

- Sinais Pares são simétricos em relação ao eixo y
- Nos sinais Pares : x(t)=x(-t)



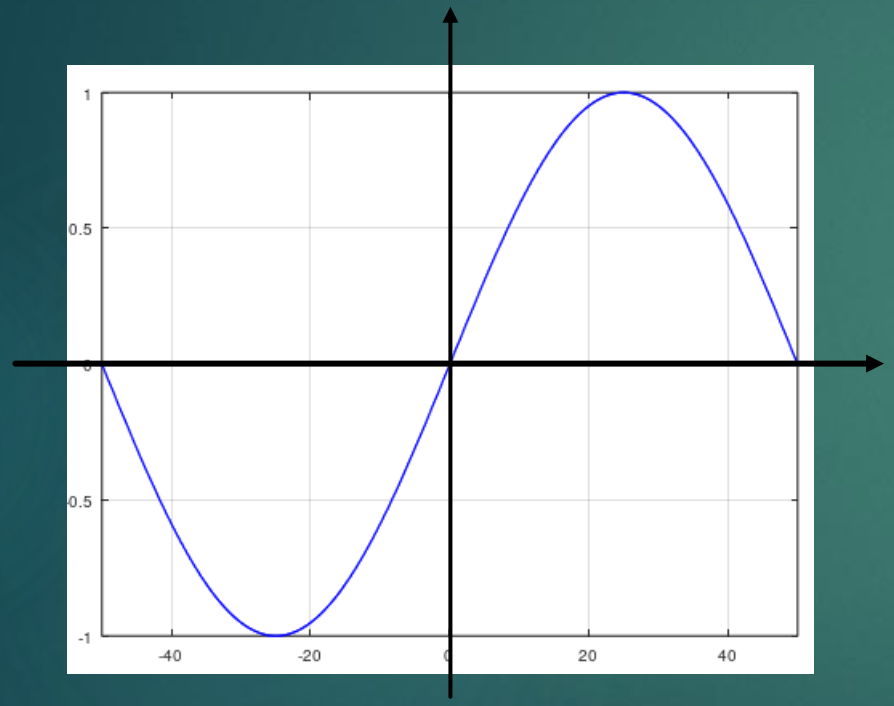

```
>> x_c = @(t) cos(2*pi*(1.6/60)*t);
>> t=linspace(-30,30,1001);
>> plot(t,x_c(t),'b');
>> grid on;
```



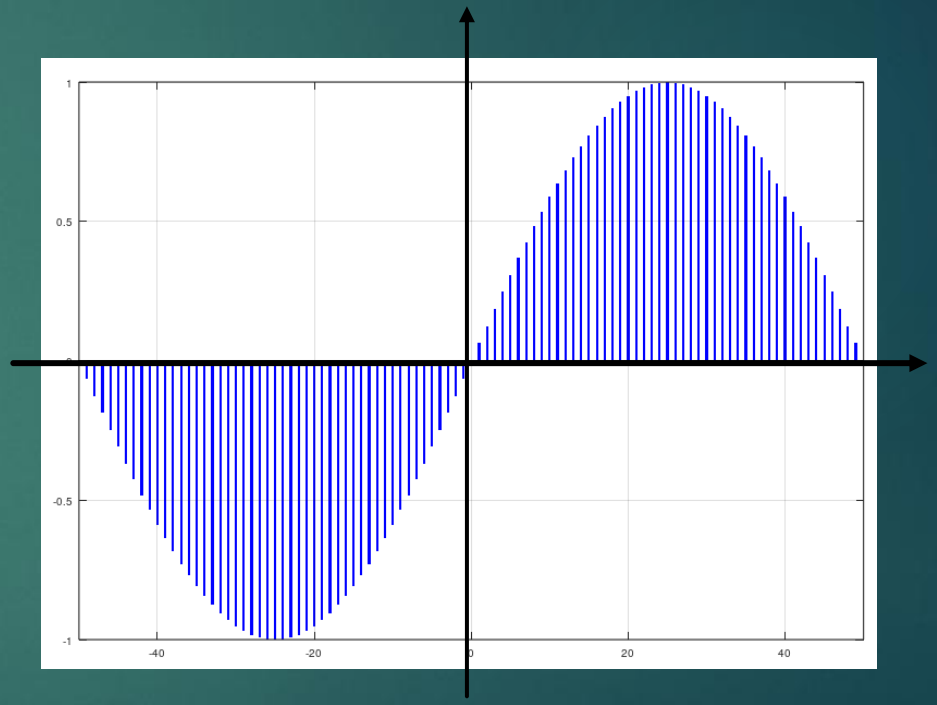

```
>> x_c = @(t) cos(2*pi*(1.6/60)*t);
>> n=[-30:30];
>> bar(n,x_c(n),0.3,'b','edgecolor','none');
>> axis([min(n),max(n)])
```

Sinais Ímpares

- Sinais ímpares são antissimétricos em relação ao eixo y.
- Nos sinais ímpares $x(t) = -x(-t)$




```
>> x_c = @(t) sin(2*pi*(1/100)*t);
>> t=linspace(-50,50,1001);
>> plot(t,x_c(t),'b');
>> grid on;
>> axis([min(t),max(t)])
```




```
>> x_c = @(t) sin(2*pi*(1/100)*t);
>> n=[-50:50];
>> bar(n,x_c(n),0.3,'b','edgecolor','none');
>> grid on;
>> axis([min(n),max(n)])
```

Decomposição de sinais nas componentes Pares e Ímpares

- $Par[x(t)] = x_e(t) = \frac{x(t)+x(-t)}{2}$

A operação Par(x) extrai a fração par de um sinal complexo.

Notar semelhança com fórmula de Euler para cos(x)

- $\text{Ímp}ar[x(t)] = x_o(t) = \frac{x(t)-x(-t)}{2}$

A operação Ímpar(x) extrai a fração ímpar de um sinal complexo.

Notar semelhança com fórmula de Euler para sin(x)

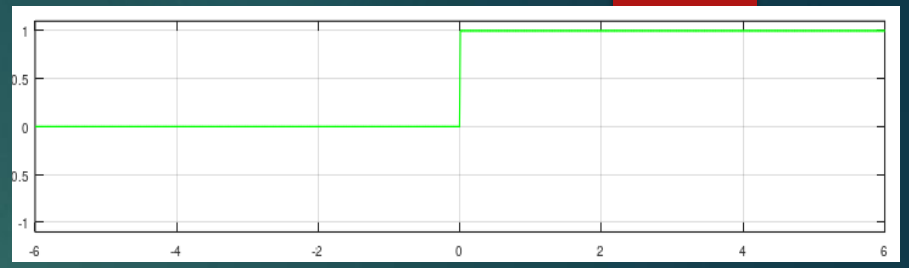
- $x(t) = x_e(t) + x_o(t)$

'e' de 'even', e 'o' de 'odd'

Exemplo – Degrau Unitário

- $x(t) = u(t)$

```
>> clf
>> plot(t,x,'g');
>> axis([min(t),max(t),-1.1,1.1])
```




```
>> subplot(311)
>> xe=(f_u(t)+f_u(-t))/2;
>> plot(t,xe,'b');
>> axis([min(t),max(t),-1.1,1.1])
>> subplot(312)
>> xo=(f_u(t)-f_u(-t))/2;
>> plot(t,xo,'r');
>> axis([min(t),max(t),-1.1,1.1])
>> subplot(313)
>> x=xe+xo;
>> plot(t,x,'k');
```

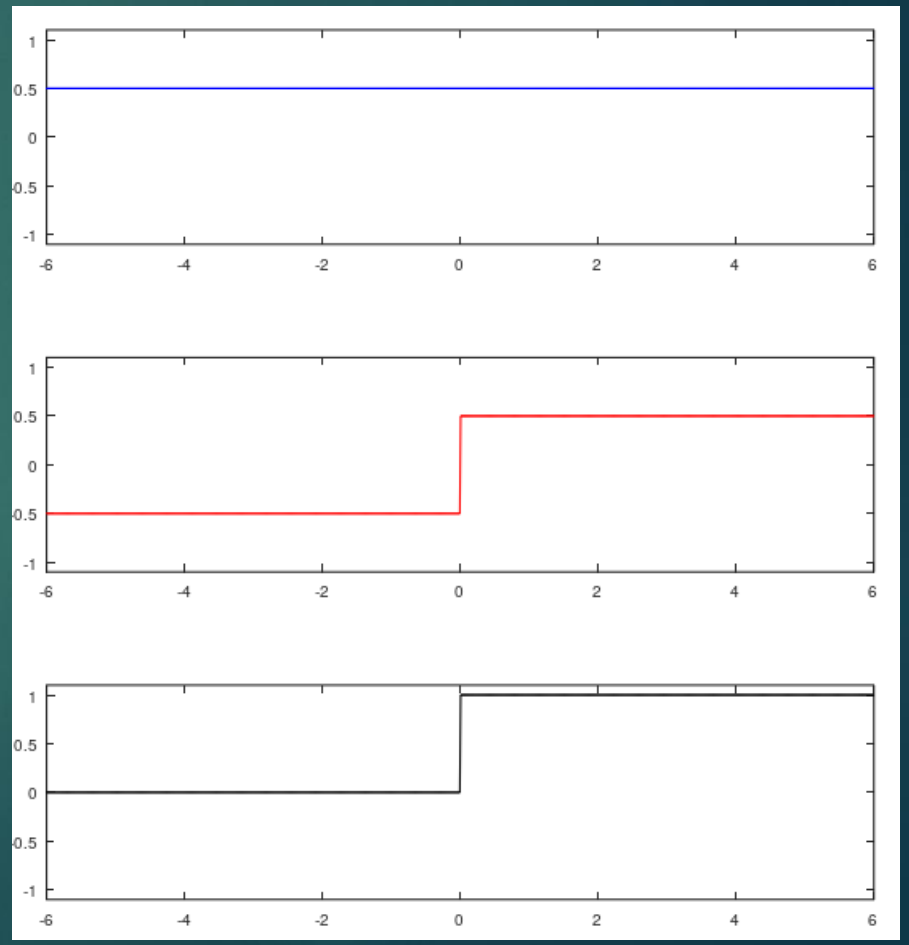
$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$+$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

$$=$$

$$x(t) = x_e(t) + x_o(t)$$



Sinais Periódicos x Sinais Aperiódicos

- Domínio contínuo

$x(t) = x(t + mT); \quad para\ todo\ t; \quad m \in \mathbb{Z}$

O período fundamental T_o é o menor T tal que $x(t) = x(t + T)$ (notar que 2 T_o satisfaria esta equação!!)

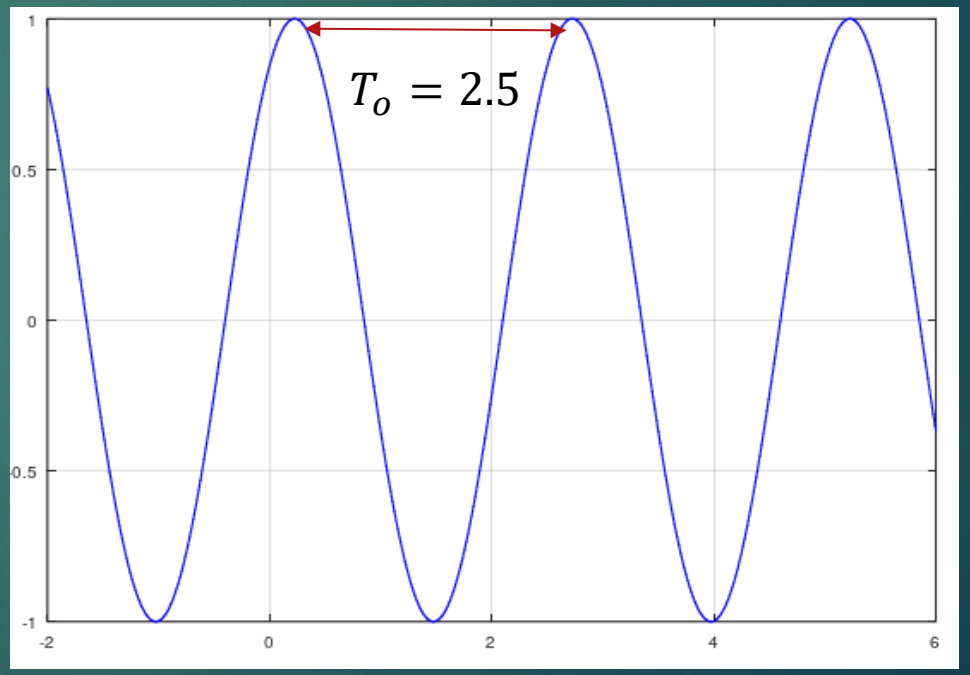
$x(t) = x(t + T_o);$

- Exemplo : $\cos\left(\frac{4\pi}{5}t + 12\right)$

$$w = frequência\ angular = \frac{4\pi}{5}\ rad/s$$

$$T_o = \frac{2\pi}{w} = \frac{2\pi}{\frac{4\pi}{5}} = \frac{10}{4} = 2.5\ \mathrm{s}$$

$$f = frequência\ em\ Hz = \frac{w}{2\pi} = \frac{1}{T_0}$$




```
>> x_c = @(t) cos(4*pi/5*t+12);
>> t=linspace(-2,6,1000);
>> plot(t,x_c(t),'b');
>> grid on;
```

Sinais Periódicos x Sinais Aperiódicos

- Domínio Discreto

$x[n] = x[n + kN]; \;\; k \in \mathbb{Z} \;\; e \;\; N \in \mathbb{Z}$

O período fundamental N_o é o menor N, tal que $x[n] = x[t + N_o]$;

(notar que 2 N_o satisfaria a equação acima para todo k!!)

- Exemplo : $\cos[wn] = \cos[w * (n + N)]$

Qualquer que seja N e qualquer que seja Z

N é um número inteiro conhecido como período.

- A periodicidade discreta ocorrerá quando existir um numero inteiro k tal que

$$wN = 2k\pi$$

O menor período será:

$$N = \min\left(\frac{2\pi}{w}k\right)$$

Exemplo 1: Periodicidade Discreta

Determinar se as seguintes funções são periódicas, e caso afirmativo, determinar seu período N_o:

$$x[n] = \cos\left(\frac{12\pi}{5}n\right)$$

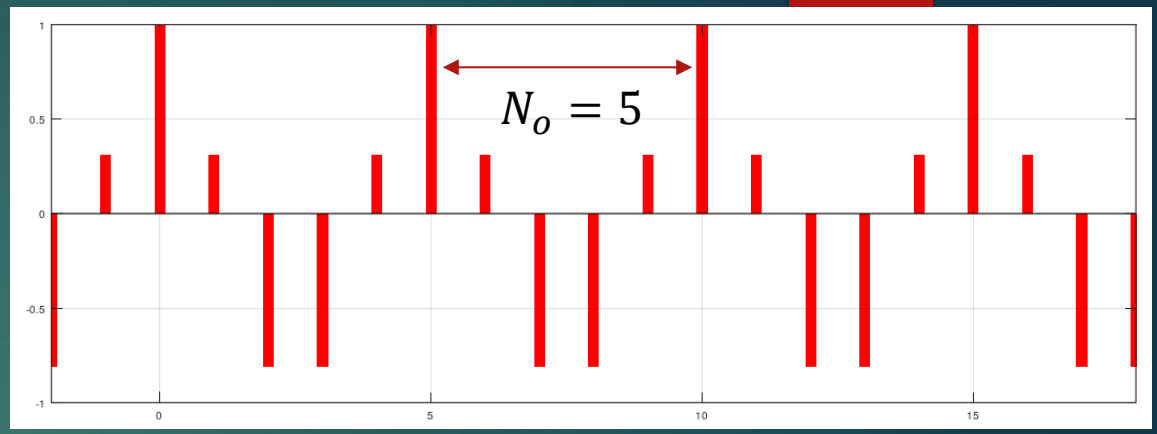
$$\frac{12\pi}{5}(n+N) = \frac{12\pi}{5}n + 2k\pi$$

$$\frac{12\pi}{5}N = 2k\pi \Rightarrow N = \frac{5k}{6}$$

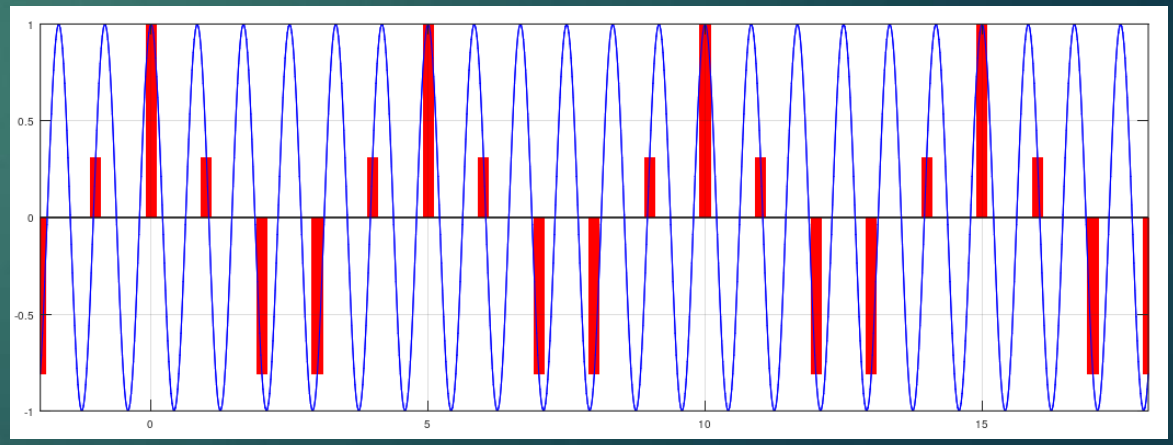
O menor N inteiro ocorre quando k=6

Então $N_o = 5$

```
>> x_c = @(t) cos(12*pi/5*t);
>> n=[-2:18];
>> bar(n,x_c(n),0.1,'r');
>> grid on;
```




```
>> x_c = @(t) cos((12*pi/5)*t);
>> t=linspace(-2,18,1001);
>> n=[-2:18];
>> hold on; grid on;
>> plot(t,x_c(t),'b');
>> bar(n,x_c(n),0.2,'r','edgecolor','none');
>> grid on;
>> axis([min(n),max(n)])
```



Exemplo 2: Periodicidade Discreta

Determinar se as seguintes funções são periódicas, e caso afirmativo, determinar seu período N_o:

$$x[n] = 5\,e^{j\frac{12\pi}{71}n} = 5cos\left(\frac{12\pi}{71}n\right) + j5sen\left(\frac{12\pi}{71}n\right)$$

$$|x[n]| = 5 \;\; \forall \;\; n$$

$$Fase(x[n]) = \frac{12\pi}{71}n$$

$$\frac{12\pi}{71}(n+N) = \frac{12\pi}{71}n + 2k\pi$$

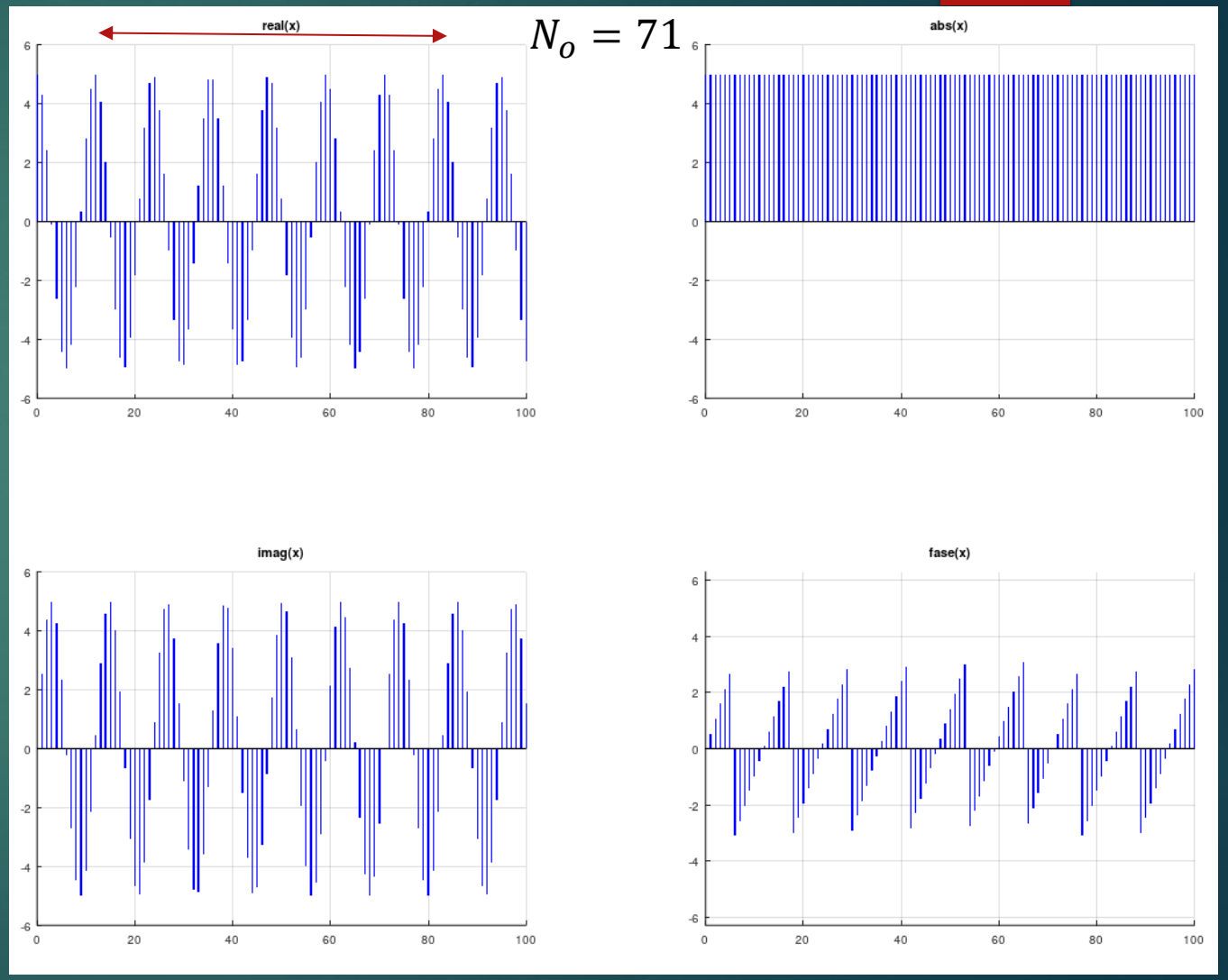
$$\frac{12\pi}{71}N = 2k\pi \Rightarrow N = \frac{71k}{6}$$

O menor N inteiro ocorre quando k=6

Então N_o =71

```
>> x_n = @(n) 5*exp((i*12*pi/71)*n);
>> n=[0:100];
>> plot_sinal_complexo_d(x_n,n)
```

$N_o = 71$

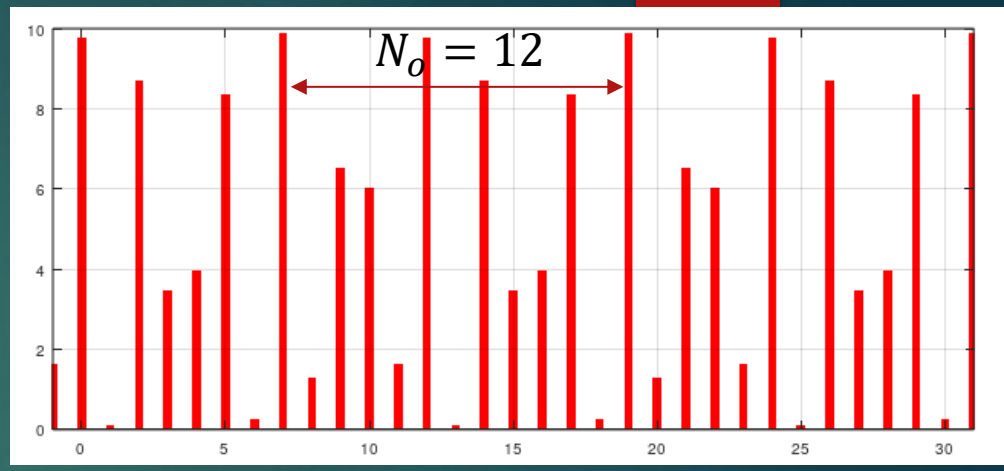


Exemplo 3: Periodicidade Discreta

Determinar se as seguintes funções são periódicas, e caso afirmativo, determinar seu período N_o:

$$x[n] = 10\,sin^2\left(\frac{7\pi}{12}n + \sqrt{2}\right)$$

```
>> x_n = @(n) 10*sin((7*pi/12)*n + sqrt(2)).^2;
>> n=[-1:31];
>> bar(n,x_n(n),0.3,'r','edgecolor','none');
>> grid on;
>> axis([min(n),max(n)])
```



$$\frac{7\pi}{12}(n+N) = \frac{7\pi}{12}n + k\pi$$

notar que soma-se $k\pi$ e não $2k\pi$,

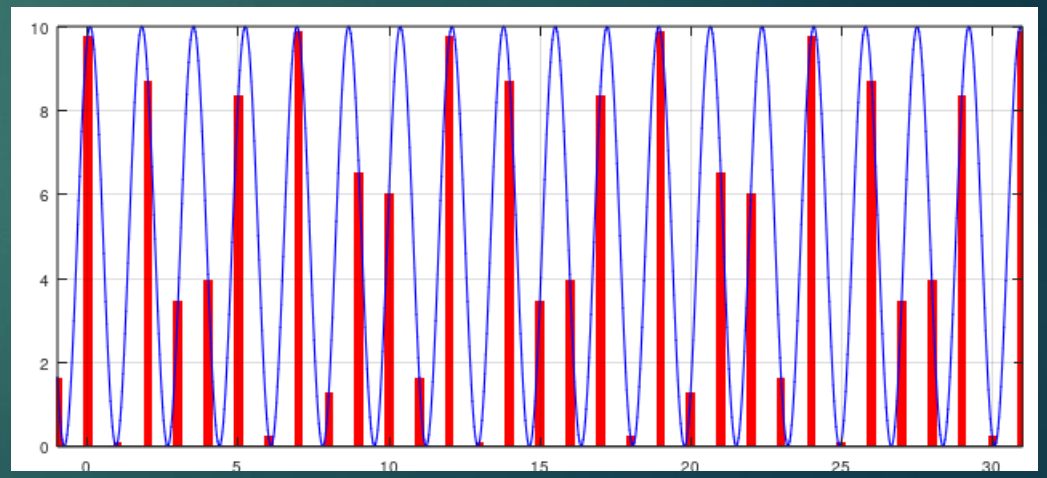
pois $sin^2(x)$ tem o dobro da frequência de sin(x)

$$\frac{7\pi}{12}N = k\pi \Rightarrow N = \frac{12k}{7}$$

O menor N inteiro ocorre quando k=7

Então $N_o = 12$

```
>> hold on; grid on;
>> t=linspace(-1,31,1001);
>> plot(t,x_n(t),'b');
```



Exemplo 4: Periodicidade Discreta

Determinar se as seguintes funções são periódicas, e caso afirmativo, determinar seu período N_o:

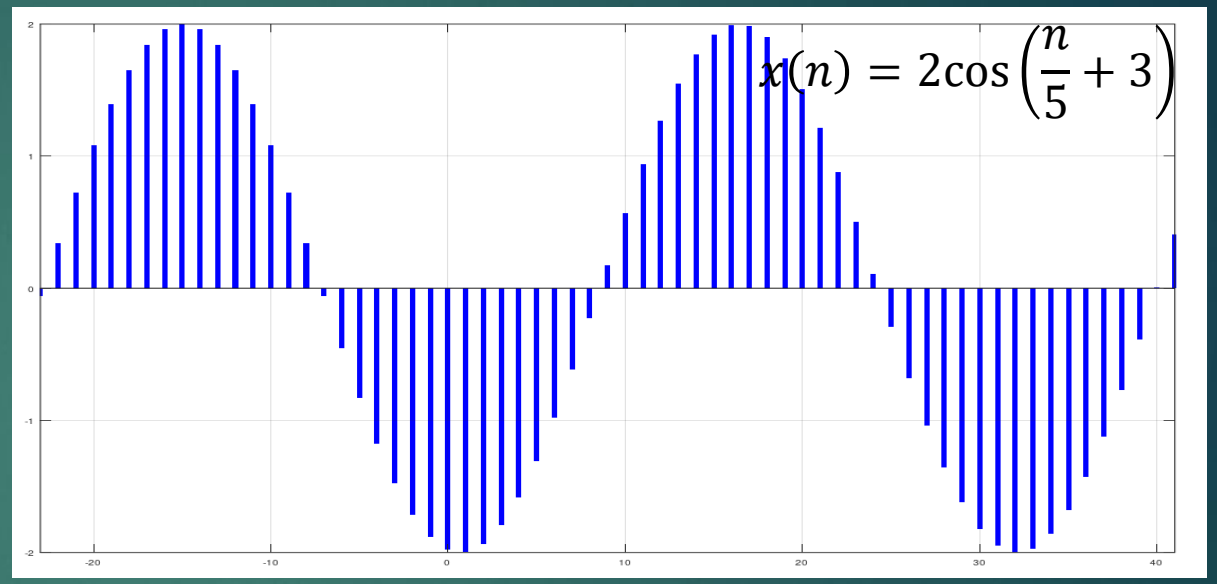
$$x[n] = 2\cos\left(\frac{n}{5} + 3\right)$$

$$\frac{1}{5}(n + N) = \frac{1}{5}n + 2k\pi$$

$$\frac{N}{5} = 2k\pi \Rightarrow N = 10k\pi$$

Nenhum k tornará N inteiro!

Então x(n) não é periódica




```
>> x_n = @(n) 2*cos( (1/5)*n + 3);
>> n=[-23:41];
>> bar(n,x_n(n),0.3,'b','edgecolor','none');
>> grid on;
>> axis([min(n),max(n)])
```

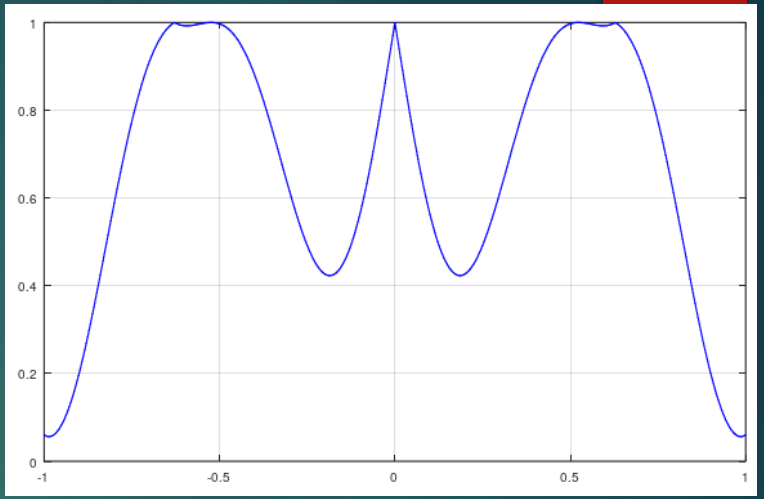
Notar que a sequência discreta x[n] não se repete exatamente após um ciclo do cosseno!!

Sinais determinísticos x Aleatórios

- Os sinais determinísticos podem ser modelados por funções matemáticas. Podemos então determinar exatamente o valor de x(t) para um dado 't'.

Exemplo: $x(t) = 1.0 - |\sin(5t)\, cos^2(3t)|$;

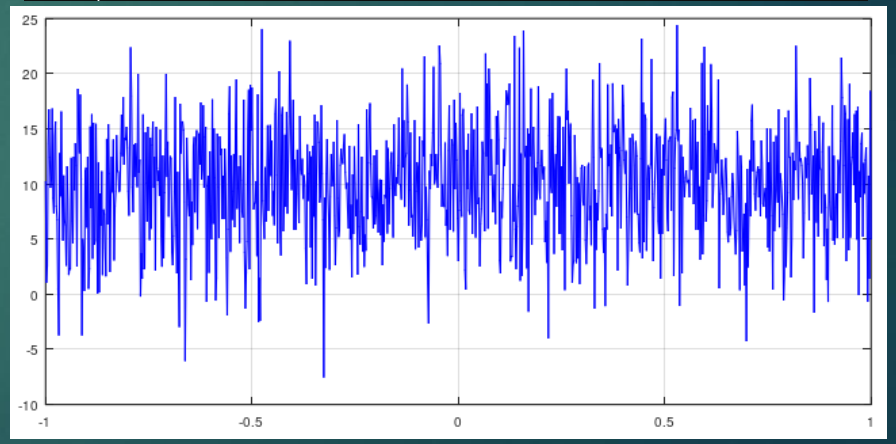
$-1 < t < 1$




```
>> t=linspace(-1,1,1001);
>> x_c = @(t) 1-abs(sin(5*t).*cos(3*t).^2);
>> plot(t,x_c(t),'b');
>> grid on;
```

- Os sinais aleatórios só podem ser caracterizados por parâmetros estatísticos. Não podemos então determinar o valor exato de r(t) para um dado 't', mas podemos dizer que r(t) vai seguir uma certa distribuição estatística.

Exemplo: r(t) é uma variável aleatória com distribuição normal, média 10.0 e desvio padrão 5.0.



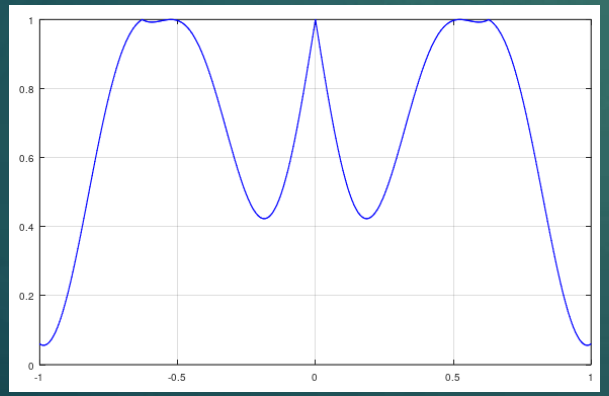

```
>> t=linspace(-1,1,1001);
>> r=10+5*randn(1,length(t));
>> plot(t,r,'b');
>> grid on;
```

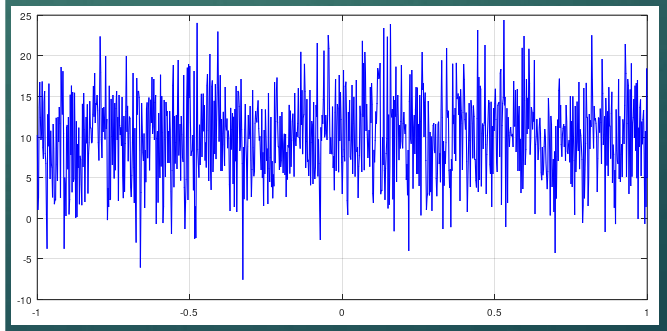
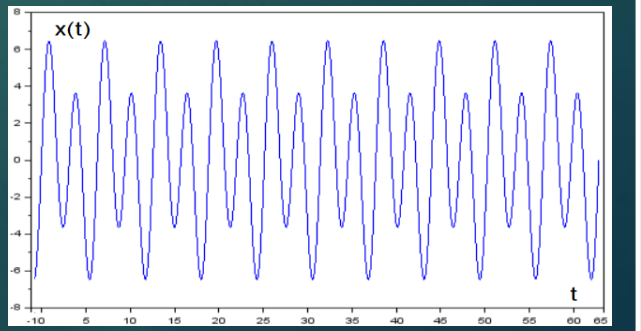
Sinais de Energia X Sinais de Potência

- $E = \int_{-\infty}^{\infty} x^2(t)dt$

$$E = \sum_{n=-\infty}^{\infty} x^2[n]$$

- Sinal de Energia
- $0 \leq E \leq \infty \quad (P \to 0)$
- Sinais determinísticos e não periódicos são normalmente sinais de energia.
- Pulsos são sinais de energia



- P= $\lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)dt$
- Se x(t) for periódico: P $= \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)dt$

- P= $\lim_{N\to\infty} \frac{1}{2N} \int_{-N}^{N} x^2[n] \quad (E \to \infty)$
- Se x(n) for periódico: P $= \frac{1}{N} \int_{0}^{N-1} x^2[n]$
- Sinal de Potência
- $0 \leq P \leq \infty$
- Sinais aleatórios e sinais periódicos são normalmente de potência.